

# EMULATOR-ASSISTED CALIBRATION OF A SEMI-ANALYTIC GALAXY FORMATION MODEL FOR THE ROMAN GALAXY REDSHIFT SURVEY

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## ABSTRACT

Realistic simulations of the *Roman* Galaxy Redshift Survey require galaxy populations with self-consistent stellar masses, star-formation rates, sizes, metallicities, dust attenuation, and emission-line luminosities. For this purpose, we present an emulator-assisted calibration of the semi-analytic model GALACTICUS. Because direct posterior exploration with GALACTICUS is computationally prohibitive, we train Gaussian-process emulators to predict the observables entering the calibration likelihood. We calibrate the model simultaneously to stellar mass functions, the star-formation-rate function, size-mass relations, the black-hole mass-velocity-dispersion relation, the mass-metallicity relation, and H $\alpha$  luminosity functions. Combining these observables breaks parameter degeneracies present in individual fits and can reveal tensions where different observables favour different regions of parameter space. The resulting model provides a broadly successful simultaneous description of the calibration data, including the abundance and redshift evolution of H $\alpha$  emitters relevant to the *Roman* Galaxy Redshift Survey. A direct GALACTICUS calculation at the maximum a posteriori point gives similarly good agreement with the calibration data. More generally, emulation makes multi-observable posterior calibration practical for computationally expensive galaxy-formation models, with the same framework readily adaptable to different observational constraints and scientific applications.

## 1. INTRODUCTION

The Nancy Grace Roman Space Telescope will carry out a wide-field near-infrared grism survey over cosmological volumes, providing galaxy redshifts for large-scale structure measurements and a large sample of emission-line galaxies for studies of galaxy evolution (Spergel et al. 2015; Wang et al. 2022). These measurements will extend galaxy redshift surveys to large samples at redshifts where the expansion history and growth of structure are less tightly constrained by existing galaxy surveys. This is especially timely given recent DESI baryon-acoustic-oscillation (BAO) measurements. DESI first-year BAO data are consistent with  $\Lambda$ CDM on their own, but when combined with CMB and Type Ia supernova data they show a mild preference for time evolution of the dark energy equation of state (DESI Collaboration et al. 2025). This is not yet conclusive evidence for evolving dark energy, but it does sharpen the case for independent measurements of the distance-redshift relation and growth history. *Roman* is valuable in this context because it will probe these quantities at high redshift, with a different instrument, galaxy selection, and set of observational systematics from existing ground-based spectroscopic surveys.

Unlike targeted spectroscopic surveys such as DESI (DESI Collaboration et al. 2016), *Roman* will use slitless spectroscopy, dispersing the light from the full field of view. This is a powerful feature of the survey, because it avoids a hard preselection of targets and can in principle provide spectra for all sufficiently bright sources in the field. It also makes the data more complicated to

interpret. Because the spectra are dispersed images of the galaxies, spatial and spectral information are mixed together, spectra from neighbouring objects can overlap, and the detectability of an emission line depends not only on its total flux but also on the size and morphology of the galaxy.

The redshifts used for the *Roman* grism survey will be obtained primarily from strong rest-frame optical emission lines, especially H $\alpha$  and [O III], with [O II] also relevant at higher redshift. For *Roman* mock catalogues, the line luminosities must be connected to the underlying galaxy population: stellar masses, star formation rates, gas metallicities, dust attenuation, galaxy sizes, and host halo environments. These correlations matter both for cosmological analyses, where the number density and bias of selected tracers are important, and for image-level simulations, where source sizes, line fluxes, continuum emission, and blending determine whether a redshift can be recovered.

There are several possible ways to build such mocks. Empirical models such as UNIVERSEMACHINE (Behroozi et al. 2019) provide powerful frameworks for connecting galaxies to dark matter haloes, while newer approaches such as DIFFSKY<sup>1</sup> extend this framework to star-formation histories and spectral-energy-distribution modelling (Alarcon et al. 2023; Hearin et al. 2023). Semi-analytic models offer a complementary approach, evolving galaxies along dark matter halo merger trees using approximate, physically motivated prescriptions for baryonic processes (e.g. Baugh 2006; Somerville & Davé 2015). They are far less computationally expensive than

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<sup>1</sup> <https://diffsky.readthedocs.io/en/latest/>

hydrodynamical simulations, but still explicitly model processes including gas cooling, star formation, feedback, metal enrichment, galaxy sizes, black-hole growth, and nebular emission. For *Roman* mock catalogues, this provides a natural way to predict many correlated properties of each galaxy within a single model.

The flexibility of semi-analytic models also makes their calibration a non-trivial part of the modelling problem. Their physical prescriptions contain parameters that are only weakly constrained by first-principles arguments, and different combinations of parameters can produce similar agreement with a small set of observables. Automated calibration methods have therefore become increasingly important. These include methods that sample the posterior distribution of model parameters (e.g. Henriques et al. 2009; Lu et al. 2011; Henriques et al. 2015), as well as optimization methods designed to search efficiently for well-fitting regions of parameter space (e.g. Ruiz et al. 2015). Comparisons in which multiple semi-analytic models are calibrated against a common set of observations have also provided insight into the successes and limitations of different models (e.g. Knebe et al. 2018).

Although semi-analytic models are computationally inexpensive compared with hydrodynamical simulations, evaluating them directly for the large number of parameter combinations required for posterior inference remains prohibitively expensive. One solution is to construct an *emulator*: a statistical model trained on a finite set of direct model evaluations that predicts the model output at new points in parameter space at much lower computational cost. Several studies have used such emulators to explore or calibrate semi-analytic models. For GALFORM, Bower et al. (2010) combined emulation with so-called *history matching* to identify the region of a 16-dimensional parameter space compatible with local luminosity functions, an approach later applied to the evolving stellar mass function by Rodrigues et al. (2017). More recently, Elliott et al. (2021) used deep-learning emulators to calibrate GALFORM to a broad set of low-redshift observables, while Madar et al. (2024) applied a related approach to H $\alpha$  emitters for *Euclid* and *Roman* survey predictions.

Our previous calibration of GALACTICUS used a different strategy to reduce the computational cost (Robertson & Benson 2026). Rather than evolving a large ensemble of merger trees for every likelihood evaluation, we calibrated the model using small samples of galaxies at a few fixed halo masses. This approach is particularly well suited to quantities for which there are observational constraints at fixed halo mass, with stellar mass being the clearest example (e.g. observationally inferred stellar-to-halo mass relations from Leauthaud et al. 2012; Behroozi et al. 2019). The resulting calibration produced a good match to the observed stellar mass function. However, matching stellar mass functions at several redshifts primarily constrains the total amount of star formation between those redshifts, and does not ensure that this star formation is distributed correctly among galaxies. The global star-formation history can therefore be approximately correct while the distribution of galaxy star-formation rates is not. Consequently, the model in Robertson & Benson (2026) reproduced stellar mass functions more successfully than H $\alpha$  luminosity

functions.

For the *Roman* grism survey, this distinction is important because the redshift sample is built from emission-line detections. In this work we therefore use full GALACTICUS runs to calculate the observables entering the likelihood, including emission-line luminosity functions, and make the resulting posterior inference computationally feasible using Gaussian-process emulators. We calibrate simultaneously to a broad set of observables that probe different aspects of the galaxy formation model, including stellar masses, star-formation rates, galaxy sizes, metallicities, black-hole masses, and H $\alpha$  luminosities. We focus primarily on low- and intermediate-redshift calibration data, where the observational constraints are generally better established, while also including H $\alpha$  luminosity functions at higher redshift because of their direct relevance to the *Roman* grism survey.

The emulators are trained on a finite set of full GALACTICUS calculations and provide both predictions and associated emulator uncertainties, which we take into account when comparing the model predictions with observational data. We test their accuracy using held-out GALACTICUS runs, use them to sample the posterior distribution of the model parameters, and finally verify the resulting calibrated model with a direct GALACTICUS calculation. The resulting calibration is intended both for *Roman* grism mock generation and for broader applications requiring physically motivated, self-consistent galaxy catalogues.

The paper is organized as follows. Section 2 describes the GALACTICUS model, the varied parameters, and the training campaign. Section 3 describes the emulator construction and validation. Section 4 describes the likelihood and MCMC inference. Section 5 presents the calibration results, including diagnostic subset fits and the final joint calibration. Section 6 discusses the interpretation and limitations of the calibration. We summarize in Section 7.

## 2. MODEL AND EMULATOR TRAINING SET

### 2.1. The GALACTICUS Model

We use GALACTICUS, a semi-analytic model of galaxy formation (Benson 2012). For a given dark matter halo accretion and merger history, encoded in the so-called *merger tree* (Lacey & Cole 1993), GALACTICUS follows the formation and evolution of the galaxies associated with each halo and subhalo. The calculation includes parameterised prescriptions for gas cooling, star formation, stellar feedback, AGN feedback, galaxy sizes, galaxy mergers, black-hole growth, and metal enrichment.

The fixed modelling choices used here are mostly the same as in Robertson & Benson (2026), while we vary a substantially larger set of numerical parameters within this model, as described in Section 2.3. We therefore focus here on the fixed modelling choices that differ from Robertson & Benson (2026) and on the parameters varied during the calibration. The GALACTICUS code is open source,<sup>2</sup> with detailed documentation of its physical prescriptions available online.<sup>3</sup> The complete parameter

<sup>2</sup> <https://github.com/galacticusorg/galacticus>

<sup>3</sup> <https://galacticus.readthedocs.io/en/stable/index.html>

files used in this work have been made publicly available alongside the paper, providing the full specification of the model configuration used here.<sup>4</sup>

The main fixed model choice that differs from [Robertson & Benson \(2026\)](#) is the treatment of galaxy sizes. In that work, galaxy component sizes were computed using an iterative equilibrium calculation, in which the baryons contribute to the gravitational potential used to determine component radii. In the larger-volume runs needed here, this produced a small sub-population of galaxies with unrealistically small sizes. We therefore use a simpler size calculation, in which component radii are determined from their specific angular momentum in the dark matter halo potential, without allowing the baryons to modify that potential.<sup>5</sup>

## 2.2. Merger Trees And Halo Sampling

For the training runs, GALACTICUS is run on extended Press–Schechter merger trees following the method of [Parkinson et al. \(2008\)](#). We sample 17,385  $z = 0$  host haloes over the mass range  $10^{11} < M_{\text{halo}}/M_{\odot} < 3 \times 10^{14}$  from a Sheth–Tormen halo mass function ([Sheth & Tormen 1999](#)).<sup>6</sup> Each  $z = 0$  host halo contains a central galaxy and may also host satellite galaxies. The calculation follows the complete merger tree of each host, including all progenitor branches. The abundance of host haloes over this mass range corresponds to an effective comoving volume of  $V_{\text{eff}} \simeq 1.3 \times 10^6 \text{ Mpc}^3$ , equivalent to a cubic box with side length  $\simeq 109 \text{ Mpc}$ . Because the calibrated model is intended to generate *Roman* mocks using the UNIT simulations ([Chuang et al. 2019](#)), we use a merger-tree mass resolution of  $2 \times 10^{10} M_{\odot}$ , similar to the minimum halo mass resolved in UNIT. For the same reason, the merger trees are rebinned onto the UNIT simulation output times, following the approach of [Benson et al. \(2012\)](#).

The use of extended Press–Schechter trees provides direct control over both the number of simulated haloes and their mass distribution, allowing us to obtain well-sampled predictions for the different calibration observables. We verify in Section 5.3 that the calibrated model gives consistent predictions when applied instead to merger trees from the UNIT  $N$ -body simulation.

## 2.3. Calibrated Galacticus Parameters

Our aim is to calibrate a model suitable for mock-catalogue generation, for which accurately reproducing a broad range of observed galaxy properties is more important than reserving observables as independent tests of the model. We therefore vary 20 GALACTICUS parameters and constrain them using several galaxy observables simultaneously.

This differs from the strategy often adopted for cosmological hydrodynamical simulations, where a relatively small set of observables is used for calibration so that other galaxy properties can serve as independent tests of

the model (e.g. [Schaye et al. 2015](#); [Nelson et al. 2018](#); [Schaye et al. 2026](#)). Our goal here is instead to obtain a model that simultaneously reproduces the range of galaxy properties required for the intended applications.

Our 20 free parameters control aspects of gas cooling, star formation, stellar feedback, black-hole seeding and growth, AGN feedback, reincorporation of ejected gas, the classification of major and minor mergers and the sizes of merger remnants, angular-momentum transfer during disk instabilities, and the stellar metal yield. The parameters and their adopted ranges are listed in Table 1. The priors are intentionally broad and are generally centred on values used in previous GALACTICUS configurations or suggested by earlier calibration experiments, rather than representing precise prior constraints on the underlying physical parameters.

## 2.4. Emulator Training Set

The emulator training set consists of  $N_{\text{train}} = 1024$  GALACTICUS model evaluations, with parameter values drawn from a scrambled Sobol sequence ([Sobol’ 1967](#); [Owen 1997](#)).<sup>7</sup> Compared with random sampling, this provides a more uniform coverage of parameter space, reducing the likelihood of large unsampled regions and generally improving emulator accuracy for a fixed number of model evaluations. We generate Sobol points in the unit hypercube and transform each coordinate through the inverse cumulative distribution function of the corresponding parameter prior. As a result, the training density follows the prior density specified in Table 1, while retaining the space-filling properties of the Sobol design in prior-quantile space. The choice of 1024 training evaluations represents a compromise between reducing emulator error across the 20-dimensional parameter space and keeping the computational cost of the GALACTICUS runs manageable. Section 3.3 uses the low-redshift stellar mass function as a concrete example to illustrate how emulator accuracy changes with training-set size.

## 2.5. Emission-Line And Dust Post-Processing

A model that produces realistic emission-line luminosities is required for the *Roman* grism-survey application that motivates this calibration, because the *Roman* redshift sample will be selected primarily through rest-frame optical emission lines. We therefore include the  $\text{H}\alpha$  luminosity functions of [Sobral et al. \(2013\)](#) among the calibration observables, requiring predictions for both intrinsic  $\text{H}\alpha$  emission and its attenuation by dust.

The  $\text{H}\alpha$  emission associated with star formation is produced mainly by recombination in  $\text{H II}$  regions around young massive stars, and therefore traces recent star formation (e.g. [Kennicutt 1998](#)). We calculate this emission using the GALACTICUS implementation described by [Merson et al. \(2018\)](#), which uses pre-computed CLOUDY ([Ferland et al. 2013](#)) photoionization models together with the star-formation and chemical-enrichment histories of each galaxy. For each GALACTICUS run, we use the resulting intrinsic  $\text{H}\alpha$  luminosities stored in the output catalogue, combining the contributions from star-forming regions in the disk and spheroid with any contribution from the AGN.

<sup>4</sup> [TODO: Add repository/version and parameter-file URL or Zenodo DOI.]

<sup>5</sup> Specifically, [Robertson & Benson \(2026\)](#) used `galacticStructureSolverEquilibrium`, whereas here we use `galacticStructureSolverSimple`.

<sup>6</sup> With  $a = 0.791$ ,  $p = 0.218$ , and normalisation 0.302, the parameter values from [Benson et al. \(2019\)](#)

<sup>7</sup> The design was generated with `scipy.stats.qmc.Sobol`, using `scramble=True`, `seed=42`, and `random_base2(m=10)`.

**Table 1**

Parameters varied in the emulator training campaign. Here  $\mathcal{LN}_{[a,b]}(x_0, \sigma_{\ln})$  denotes a log-normal prior truncated to  $[a, b]$ , with median  $x_0$  and logarithmic width  $\sigma_{\ln}$ , while  $\mathcal{U}_{[a,b]}$  denotes a uniform prior. Units for dimensional parameters are shown in the parameter column, and priors are given for the numerical value in those units.

| Parameter                                     | Description                                                                              | Prior                                       |
|-----------------------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Star formation and stellar populations</b> |                                                                                          |                                             |
| $\nu_{\text{SF,d}}/\text{Gyr}^{-1}$           | Disk star-formation frequency normalization                                              | $\mathcal{LN}_{[0.1,10]}(1, 2)$             |
| $\epsilon_{*,\text{sph}}$                     | Spheroid star-formation efficiency per dynamical time                                    | $\mathcal{LN}_{[10^{-4},1]}(0.01, 4)$       |
| $Y_Z$                                         | Stellar metal yield                                                                      | $\mathcal{LN}_{[0.01,0.04]}(0.02, 0.5)$     |
| <b>Stellar feedback</b>                       |                                                                                          |                                             |
| $V_{\text{out,d}}/\text{km s}^{-1}$           | Disk stellar-feedback velocity scale                                                     | $\mathcal{LN}_{[25,300]}(150, 0.5)$         |
| $\alpha_{\text{out,d}}$                       | Disk stellar-feedback velocity exponent                                                  | $\mathcal{U}_{[0,4]}$                       |
| $V_{\text{out,sph}}/\text{km s}^{-1}$         | Spheroid stellar-feedback velocity scale                                                 | $\mathcal{LN}_{[10,150]}(50, 1)$            |
| <b>Gas cooling and recycling</b>              |                                                                                          |                                             |
| $f_{\text{cool}}$                             | Gas cooling-rate multiplier                                                              | $\mathcal{LN}_{[0.05,1.2]}(0.5, 1)$         |
| $r_{\text{core}}/R_{\text{vir}}$              | Hot-gas core radius in units of virial radius                                            | $\mathcal{LN}_{[0.03,1]}(0.3, 1)$           |
| $\gamma_{\text{reinc}}$                       | Ejected-gas reincorporation normalization                                                | $\mathcal{LN}_{[1,25]}(5, 0.5)$             |
| $\delta_{\text{reinc},1}$                     | Redshift exponent in the Henriques reincorporation model                                 | $\mathcal{U}_{[-2,4]}$                      |
| $\delta_{\text{reinc},2}$                     | Virial-velocity exponent in the Henriques reincorporation model                          | $\mathcal{U}_{[-1,5]}$                      |
| <b>Mergers and disk instabilities</b>         |                                                                                          |                                             |
| $f_{\text{major}}$                            | Mass-ratio threshold for major mergers                                                   | $\mathcal{LN}_{[0.1,0.5]}(0.25, 0.5)$       |
| $f_{\text{orbit}}$                            | Orbital-energy factor for merger-remnant sizes                                           | $\mathcal{LN}_{[0.25,4]}(1, 0.5)$           |
| $f_{j,\text{sph}}^{\text{bar}}$               | Angular-momentum fraction retained by the spheroid in bar instabilities                  | $\mathcal{LN}_{[0.05,0.95]}(0.2, 0.5)$      |
| <b>Supermassive black holes</b>               |                                                                                          |                                             |
| $M_{\text{BH,seed}}/M_{\odot}$                | Black-hole seed mass                                                                     | $\mathcal{LN}_{[100,100\,000]}(3000, 4)$    |
| $\epsilon_{\text{BH,radio}}$                  | Radio-mode black-hole heating efficiency                                                 | $\mathcal{LN}_{[0.03,1]}(0.3, 0.5)$         |
| $\epsilon_{\text{BH,wind}}$                   | Black-hole wind feedback efficiency                                                      | $\mathcal{LN}_{[0.00024,0.024]}(0.0024, 1)$ |
| $\alpha_{\text{Bondi,sph}}$                   | Bondi accretion enhancement for spheroid gas                                             | $\mathcal{LN}_{[0.05,100]}(5, 2)$           |
| $\alpha_{\text{Bondi,hot}}$                   | Bondi accretion enhancement for hot-halo gas                                             | $\mathcal{LN}_{[0.05,100]}(6, 2)$           |
| $x_{\text{thin,max}}$                         | Upper transition value for $\dot{M}c^2/L_{\text{Edd}}$ for switched thin-disk/ADAF model | $\mathcal{LN}_{[0.03,30]}(3, 2)$            |

The H $\alpha$  emission associated with star formation is produced mainly by recombination in H II regions around young massive stars, and therefore traces recent star formation (e.g. Kennicutt 1998). We calculate this emission using the GALACTICUS implementation described by Merson et al. (2018), which uses pre-computed CLOUDY (Ferland et al. 2013) photoionization models together with the star-formation and chemical-enrichment histories of each galaxy. For each GALACTICUS run, we use the resulting intrinsic H $\alpha$  luminosities stored in the output catalogue, combining the contributions from star-forming regions in the disk and spheroid with any contribution from the AGN.

These intrinsic luminosities cannot be compared directly with observed luminosity functions, because H $\alpha$  emission is attenuated by dust before escaping the galaxy. We therefore apply a dust-attenuation model in post-processing. As our baseline, we adopt the local relation between H $\alpha$  attenuation and stellar mass measured by Garn & Best (2010), which we refer to as GB10. This relation is based on Balmer-decrement measurements, which estimate dust attenuation from ratios of hydrogen recombination lines. Balmer-decrement measurements at higher redshift suggest that a mass-dependent attenuation prescription remains a reasonable baseline, although some redshift evolution is possible (e.g. Domínguez et al. 2013). We therefore allow departures from the GB10 relation in its normalization and stellar-mass dependence, together with redshift evolution and mass-dependent redshift evolution.

For each GALACTICUS training run, emission-line luminosity functions are computed after applying this dust model. The five dust parameters, which describe departures from the GB10 relation and the scatter about it, are

sampled jointly with the 20 GALACTICUS parameters in the final inference. The mathematical form of the attenuation model, its parameter priors, and the construction of the attenuated H $\alpha$  luminosity functions are described in Appendix A.

### 3. EMULATING CALIBRATION OBSERVABLES

#### 3.1. Set Of Calibration Observables

Each GALACTICUS evaluation produces a catalogue of galaxies at a set of output times. From these galaxies we compute binned observable vectors, such as stellar mass functions, luminosity functions, size–mass relations, and black hole scaling relations. The emulator is trained to reproduce these vectors as a function of the varied GALACTICUS parameters.

The final calibration uses the observable set listed in Table 2. For number-density observables we denote the binned differential number density by  $\Phi$ , with the argument indicating the quantity being binned, for example  $\Phi(M_{\star})$  for a stellar mass function and  $\Phi(L_{\text{H}\alpha})$  for an H $\alpha$  luminosity function. The emulated-quantity column of the table gives the transformed quantity passed to the emulator and used in the likelihood.

For each calibration observable, we also estimate the finite-sampling uncertainty associated with constructing the binned prediction from a finite GALACTICUS galaxy catalogue. For number-density observables, this corresponds to the counting uncertainty associated with the galaxies contributing to each bin. For binned relations, such as the size–mass, black-hole–velocity–dispersion, and mass–metallicity relations, the emulated quantity is the mean transformed galaxy property in each bin, and the finite-sampling uncertainty corresponds to the uncertainty on this mean. These per-bin uncertainties are

passed to the GP as a diagonal training-noise term, so that noisier training outputs are treated as less precise estimates of the underlying model prediction.

### 3.1.1. Observable-Space Transformations

For observables that span a large dynamic range across GALACTICUS parameter space, we emulate logarithmic quantities. For number-density observables this means emulating  $\log_{10} \Phi$  rather than  $\Phi$  itself. At fixed stellar mass, star formation rate, or emission-line luminosity, changes in the GALACTICUS parameters can produce order-of-magnitude changes in the predicted number density. The logarithmic transformation therefore gives a smoother emulator target across parameter space and makes emulator errors more closely related to fractional differences in  $\Phi$ . It also matches the space in which luminosity and mass functions are usually compared.

We use the same approach for other positive quantities with broad dynamic range. The size–mass relations are emulated as  $\log_{10} R_{\text{eff}}$  and the black-hole scaling relation as  $\log_{10} M_{\text{BH}}$ . The gas-phase mass–metallicity relation is already conventionally expressed as  $12 + \log_{10}(\text{O}/\text{H})$ , and is emulated in that form.

The transformations require special treatment when a training-set prediction is zero or undefined. For example, a model may produce no galaxies in a stellar-mass-function bin, so that  $\log_{10} \Phi$  is undefined, or no quiescent galaxies in a high-stellar-mass bin, so that the binned mean size is undefined. For number densities, we impose a small positive floor before taking the logarithm. For undefined binned relations, we replace the missing value by the median valid training-set value in that bin and assign it a large training uncertainty, so that the assigned value carries very little weight when fitting the emulator. The detailed floor and missing-value procedure is described in Appendix B.2.

### 3.1.2. Principal-Component Compression

A simple approach to emulating each observable would be to build a separate emulator for every bin: for example, one emulator for the number density of galaxies at  $M_* \simeq 10^{10} M_{\odot}$ , another at  $M_* \simeq 2 \times 10^{10} M_{\odot}$ , and so on. This is straightforward, but it treats neighbouring bins as independent targets even though changes in GALACTICUS parameters usually move the predictions coherently across an observable. For example, a change in feedback strength will typically shift the amplitude, slope, or knee of a stellar mass function, rather than perturbing each stellar-mass bin independently.

We therefore use principal-component analysis (PCA) to compress each observable vector before emulation. This reduces the number of emulators that must be trained and evaluated, and can improve predictive accuracy by exploiting correlations in how the different bins of an observable vary across parameter space. Principal-component compression is commonly used when emulating expensive simulation outputs with many correlated bins, including recent applications to cosmological and galaxy-formation inference (e.g. Ramachandra et al. 2026). For a given observable, let  $\theta$  denote the relevant model parameters: the varied GALACTICUS parameters and, where applicable, the dust-model parameters. We write

$$\mathbf{y}(\theta) = (y_1(\theta), \dots, y_{N_{\text{bin}}}(\theta)) \quad (1)$$

for the vector of predictions in the  $N_{\text{bin}}$  bins used for that observable. For the stellar mass function, for example,  $y_b(\theta) = \log_{10} \Phi_b(\theta)$ , where  $\Phi_b$  is the predicted number density in the  $b$ th stellar-mass bin. For each GALACTICUS run in the training set we have one such vector, so the training data for this observable form an  $N_{\text{train}} \times N_{\text{bin}}$  matrix; we use  $n$  to index the individual GALACTICUS runs in the training set.

We first standardise each bin using its mean,  $\mu_b$ , and standard deviation,  $s_b$ , across the training-set runs. For training-set run  $n$ , the standardised value in bin  $b$  is

$$z_{nb} = \frac{y_b(\theta_n) - \mu_b}{s_b}. \quad (2)$$

We then apply PCA to the resulting  $N_{\text{train}} \times N_{\text{bin}}$  matrix with elements  $z_{nb}$ . PCA defines an orthonormal set of directions,  $\mathbf{e}_k$ , in observable-bin space, ordered by the variance of the training set projected along each direction (e.g. Jolliffe & Cadima 2016). For training model  $n$ , its coefficient along component  $k$  is

$$a_{nk} = \mathbf{e}_k \cdot \mathbf{z}_n = \sum_b e_{kb} z_{nb}. \quad (3)$$

We retain the smallest number of components,  $K$ , required to explain 99 per cent of the variance in the standardised training set. The quantities  $a_{nk}$  are then the training targets for the emulator associated with component  $k$ ; after training, the emulator predicts the corresponding coefficient,  $a_k(\theta)$ , at new parameter values. The predicted observable vector in standardised observable space is reconstructed as

$$\mathbf{z}(\theta) \simeq \sum_{k=1}^K a_k(\theta) \mathbf{e}_k. \quad (4)$$

Finally, we undo the bin-wise standardisation to recover the emulator target in each bin,

$$y_b(\theta) \simeq \mu_b + s_b z_b(\theta). \quad (5)$$

For comparison and diagnostic plots, we also train bin-by-bin emulators. The main calibration results use PCA emulators with a 99 per cent variance threshold; the cross-validation tests below show why this is a reasonable choice. The number of retained PCA components for each calibration observable is listed in Table 2.

### 3.2. Gaussian-Process Emulators

Each retained PCA coefficient is modelled with a Gaussian process (GP) as a function of the input parameters. The same basic GP choices are used for the emulators compared below, so that the validation tests mainly probe the effects of training-set size and PCA compression. For the 20 GALACTICUS parameters, the GP inputs are not the physical parameter values themselves, but the corresponding prior quantiles. That is, for GALACTICUS parameter  $\theta_i$  with prior cumulative distribution function  $P_i(<\theta_i)$ , we train the emulator using

$$u_i = P_i(<\theta_i), \quad (6)$$

so that each input parameter is mapped onto the unit interval. This is the space in which the Sobol design was constructed, and is therefore the space in which the training data are approximately space filling. It also avoids

**Table 2**

Calibration observables used in this work. The final joint fit uses all listed observables. The emulated quantity defines the space used for emulator training and posterior inference, and is the space in which validation RMSE values are computed.  $N_{\text{bin}}$  is the number of binned data-vector elements entering the emulator and likelihood for the corresponding row.  $N_{\text{PC}}$  is the number of principal components required to explain 99 per cent of the variance in the standardised training set.

| Calibration observable                  | Reference data            | Emulated quantity                         | Redshift          | $N_{\text{bin}}$ | $N_{\text{PC}}$ |
|-----------------------------------------|---------------------------|-------------------------------------------|-------------------|------------------|-----------------|
| Stellar mass function                   | Tomczak et al. (2014)     | $\log_{10} \Phi(M_{\star})$               | $0.20 < z < 0.50$ | 14               | 6               |
|                                         |                           |                                           | $1.00 < z < 1.25$ | 11               | 4               |
| Star-formation-rate function            | Robotham et al. (2011)    | $\log_{10} \Phi(\dot{M}_{\star})$         | $0.013 < z < 0.1$ | 25               | 7               |
| Size-mass relation (star-forming)       | van der Wel et al. (2014) | $\log_{10} R_{\text{eff}}(M_{\star})$     | $0 < z < 0.5$     | 4                | 3               |
| Size-mass relation (quiescent)          |                           | $\log_{10} R_{\text{eff}}(M_{\star})$     | $0 < z < 0.5$     | 5                | 4               |
| Black-hole-velocity-dispersion relation | McConnell & Ma (2013)     | $\log_{10} M_{\text{BH}}(\sigma_{\star})$ | $z \approx 0$     | 5                | 4               |
| Gas-phase mass-metallicity relation     | Blanc et al. (2019)       | $12 + \log_{10}(\text{O}/\text{H})$       | $z \approx 0$     | 49               | 6               |
|                                         |                           |                                           | $z = 0.40$        | 18               | 6               |
|                                         |                           |                                           | $z = 0.84$        | 9                | 5               |
|                                         |                           |                                           | $z = 1.47$        | 13               | 7               |
| $\text{H}\alpha$ luminosity function    | Sobral et al. (2013)      | $\log_{10} \Phi(L_{\text{H}\alpha})$      | $z = 2.23$        | 15               | 8               |

asking the GP to learn across raw parameters with very different units, ranges, and prior shapes; an interval of fixed width in  $u_i$  contains the same prior probability for each parameter. The  $\text{H}\alpha$  luminosity-function emulator has five additional inputs: the dust post-processing parameters described in Section 2.5.

After the PCA transformation, each retained coefficient is standardised once more before fitting its GP, by subtracting its training-set mean and dividing by its training-set standard deviation. This is separate from the bin-wise standardisation applied before PCA, and is undone before reconstructing the observable vector. Let  $\mathbf{u}$  denote the complete input vector supplied to a given GP after the input transformations just described. The kernel specifies the covariance between the values of the function at two points in this input space. We use a Matérn kernel with  $\nu = 2.5$ , multiplied by a constant amplitude, and allow a separate length scale for each input coordinate. For two input vectors,  $\mathbf{u}$  and  $\mathbf{u}'$ , the kernel is

$$k(\mathbf{u}, \mathbf{u}') = A \left( 1 + \sqrt{5}r + \frac{5r^2}{3} \right) \exp(-\sqrt{5}r), \quad (7)$$

where

$$r^2 = \sum_i \left( \frac{u_i - u'_i}{\ell_i} \right)^2. \quad (8)$$

Here  $A$  is the covariance amplitude and  $\ell_i$  is the length scale associated with the  $i$ th emulator input. Small values of  $\ell_i$  allow the emulator prediction to vary rapidly as that input coordinate is changed, while large values correspond to smoother variation. For each PCA coefficient, the kernel hyperparameters are therefore the overall covariance amplitude and one length scale for each emulator input. We determine the kernel hyperparameters by maximizing the GP marginal likelihood (Rasmussen & Williams 2006) using a gradient-based optimizer, starting from multiple initial points to reduce sensitivity to local maxima.

### 3.2.1. Emulator Uncertainties

The observables measured from each GALACTICUS training run have finite-sampling uncertainties, arising from the finite number of galaxies used to estimate each

binned quantity. We account for these uncertainties during GP training by adding their variances to the diagonal of the covariance matrix obtained by evaluating the kernel at the training points (e.g. Rasmussen & Williams 2006). The training uncertainty can therefore vary between individual training models and observable bins. For PCA emulators, the per-bin uncertainties are propagated through the standardisation and PCA transformations to obtain a training uncertainty for each retained PCA coefficient. We neglect correlations between observable bins in this propagation. Further details are given in Appendix B.3.

At a new point in parameter space, the GP provides both a mean prediction and a predictive variance that quantifies the emulator's uncertainty in that prediction. We propagate this predictive variance into the calibration likelihood by combining it with the observational uncertainty (see Section 4.2). This requires the emulator uncertainties to be reasonably calibrated, making the held-out tests in Section 3.2.2 an important part of assessing the reliability of the inference.

### 3.2.2. Held-Out Emulator Validation

We validate the emulator using *five-fold cross-validation*. The GALACTICUS training runs are divided into five equally sized subsets. For each fold, the emulator is trained on four subsets and evaluated on the remaining subset, so that every GALACTICUS evaluation is predicted once by an emulator that was not trained using that model. Unless otherwise stated, all validation results in this section are based on the combined predictions from these five folds.

These held-out predictions test whether the emulator can reproduce direct GALACTICUS calculations at new parameter values, and allow us to assess how choices such as training-set size and PCA compression affect its predictive accuracy. A summary of the held-out performance for all calibration-observable emulators is given in Appendix B.1, where we also account explicitly for the finite-sampling uncertainty of the held-out GALACTICUS calculations. Section 3.3 illustrates the validation procedure for the low-redshift stellar mass function.

To quantify this comparison, we consider individual bins of the observable vector. For a given bin, we compare the emulator prediction,  $m_i$ , with the direct

GALACTICUS value,  $g_i$ , for each validation model  $i$ . The root-mean-square error,

$$\text{RMSE} = \left[ \frac{1}{N_{\text{val}}} \sum_i (m_i - g_i)^2 \right]^{1/2}, \quad (9)$$

measures the typical difference between the emulator and direct GALACTICUS predictions. We also use the coefficient of determination,

$$R^2 = 1 - \frac{\sum_i (m_i - g_i)^2}{\sum_i (g_i - \bar{g})^2}, \quad (10)$$

where  $\bar{g}$  is the mean direct GALACTICUS value over all validation models. This compares the emulator error with the variation of the direct GALACTICUS predictions in that bin:  $R^2 = 1$  corresponds to perfect predictions, while  $R^2 = 0$  corresponds to doing no better than predicting the mean GALACTICUS value for all parameter choices. For the stellar mass functions shown below, both  $m_i$  and  $g_i$  are values of  $\log_{10} \Phi$  in a fixed stellar-mass bin, so the RMSE is measured in dex.

### 3.3. Low-Redshift Stellar Mass Function Example

To give some intuition for the validation procedure and metrics discussed above, we now consider one concrete example: the low-redshift stellar mass function. We show the emulator predictions alongside the corresponding held-out GALACTICUS results, and then examine the validation metrics and their dependence on the construction and size of the training set.

Figure 1 illustrates the held-out validation procedure for the low-redshift stellar mass function. The emulator reproduces the broad variation in stellar mass functions across the held-out models, with the largest discrepancies occurring towards the high-mass end and for models with more extreme mass functions.

Figure 2 shows this bin-level comparison for three representative stellar-mass bins from Figure 1. The plotted points should lie close to the one-to-one line if the emulator is accurate. The RMSE values measure the absolute scatter about this line, while the  $R^2$  values measure how small that scatter is relative to the variation in the direct GALACTICUS predictions.

Figure 3 shows how these diagnostics depend on training-set size and emulator construction for several representative stellar-mass bins. We compare emulators trained on the first 64, 128, 256, 512, and 1024 points of the Sobol sequence. Because the first  $2^m$  points of a Sobol sequence are themselves designed to provide good coverage of parameter space, these subsets provide a natural sequence of training sets for assessing the effect of increasing the training-set size. Emulator accuracy improves substantially as the training set grows from 64 to 1024 models, although the improvement becomes smaller for the larger training sets, suggesting diminishing returns from further increasing the training-set size. We therefore adopt the 1024-run training set as a practical compromise between emulator accuracy and computational cost.

Figure 3 also compares different levels of PCA compression. Retaining 90 per cent of the PCA variance leads to somewhat poorer held-out accuracy, while the 99 and 99.9 per cent thresholds give very similar results.

Their performance is also comparable to, and in some cases slightly better than, the bin-by-bin emulators. We therefore adopt the 99 per cent threshold for the fiducial emulator, since retaining additional PCA components produces no clear improvement in predictive accuracy.

## 4. POSTERIOR INFERENCE OF MODEL PARAMETERS

### 4.1. MCMC Fits Used In This Work

We use two kinds of MCMC fit. First, we run a small set of diagnostic fits to subsets of the calibration data to illustrate how different observables constrain different combinations of GALACTICUS parameters. The example shown in Section 5.1 uses stellar mass functions, size–mass relations, and the star-formation-rate function, both separately and jointly. Second, we run the final joint calibration using the full set of calibration observables summarized in Table 2. The final emulators are trained on the 1024-run training set described in Section 2.4. The final joint fit samples the 20 GALACTICUS parameters in Table 1 together with the five dust parameters described in Appendix A.

We infer the model parameters from the posterior distribution

$$p(\boldsymbol{\theta} \mid \mathbf{d}) \propto \mathcal{L}(\mathbf{d} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}), \quad (11)$$

where  $p(\boldsymbol{\theta})$  is the product of the parameter priors specified in Table 1 and, for the joint calibration, the dust-parameter priors given in Appendix A.

### 4.2. The Likelihood

The MCMC likelihood is evaluated in the same transformed observable space in which the emulators are trained. For stellar mass functions, the star formation rate function, and the  $\text{H}\alpha$  luminosity functions, this is  $\log_{10}$  number density. For the size–mass relations and black-hole relation, the fitted quantities are already logarithmic quantities,  $\log_{10} R_{\text{eff}}$  and  $\log_{10} M_{\text{BH}}$ , respectively. For the mass–metallicity relation, the fitted quantity is  $12 + \log_{10}(\text{O}/\text{H})$ .

For each observable bin  $i$ , we assume a Gaussian likelihood in this transformed space,

$$\ln \mathcal{L} = -\frac{1}{2} \sum_i \left[ \frac{(d_i - m_i(\boldsymbol{\theta}))^2}{s_i^2(\boldsymbol{\theta})} + \ln(2\pi s_i^2(\boldsymbol{\theta})) \right], \quad (12)$$

where  $d_i$  is the observed value,  $m_i(\boldsymbol{\theta})$  is the emulator mean prediction, and the total variance is

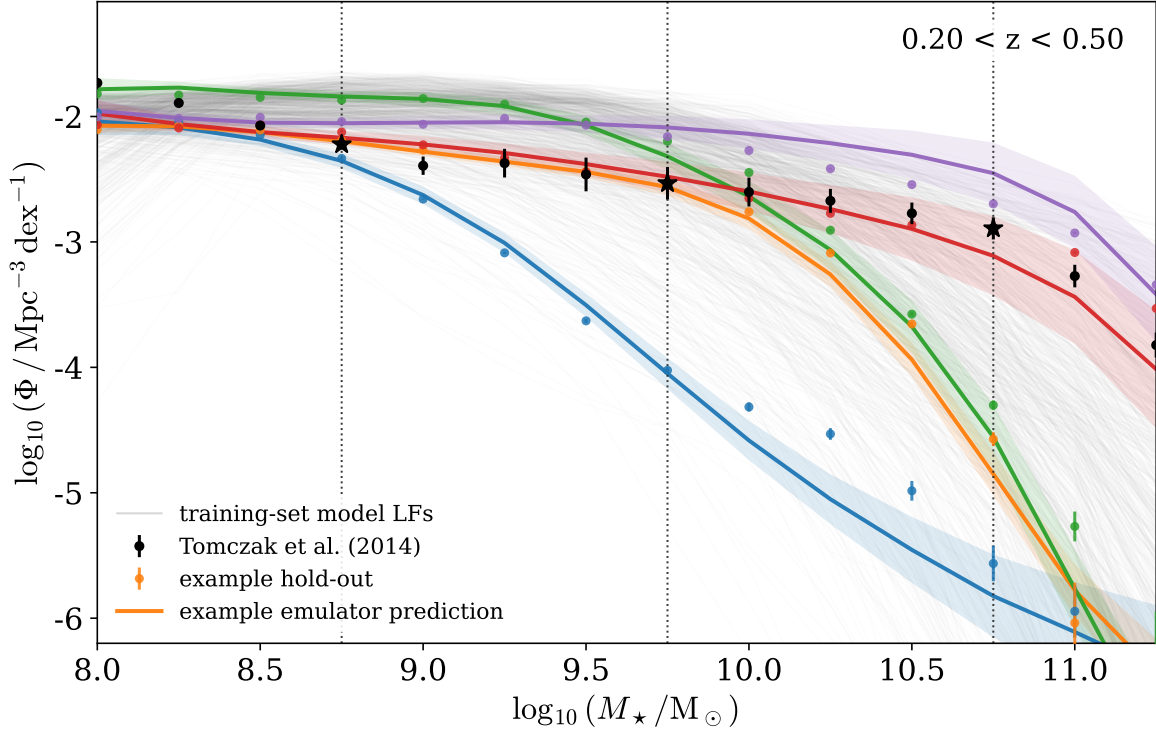
$$s_i^2(\boldsymbol{\theta}) = \sigma_{\text{data},i}^2 + \sigma_{\text{emu},i}^2(\boldsymbol{\theta}). \quad (13)$$

Here  $\sigma_{\text{data},i}$  is the observational uncertainty and  $\sigma_{\text{emu},i}(\boldsymbol{\theta})$  is the GP predictive uncertainty. The observational and emulator uncertainties are therefore added in quadrature in the transformed observable space.

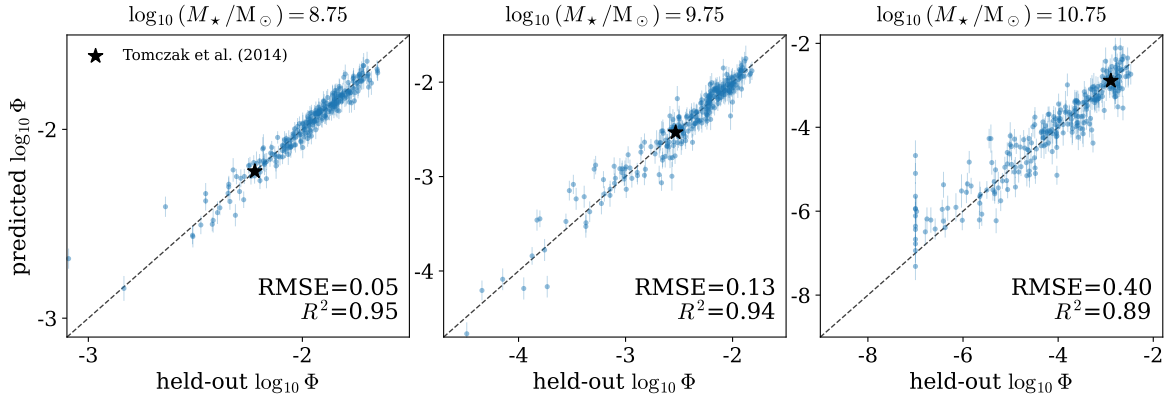
We treat the uncertainties in different observable bins as independent, using only the diagonal observational and emulator variances in the likelihood. We therefore neglect correlations both between bins of a given observable and between different calibration observables.

For number-density datasets whose uncertainties are provided in linear space, we propagate the diagonal uncertainties to logarithmic number density using

$$\sigma_{\log_{10} \Phi} \simeq \frac{\sigma_{\Phi}}{\Phi \ln 10}. \quad (14)$$



**Figure 1.** Example of held-out validation for the low- $z$  stellar mass function. Thin grey curves show the range of GALACTICUS predictions across the training set. Black points show the observed stellar mass function from Tomczak et al. (2014). Coloured points show direct GALACTICUS predictions for example held-out parameter choices. The corresponding coloured curves show emulator predictions for the same parameter choices, where in each case the emulator was trained without using the corresponding GALACTICUS evaluation. The shaded regions show the emulator uncertainty. Five held-out models spanning the range of predicted stellar mass functions are selected systematically for visual clarity; the validation metrics use the full held-out sample. Vertical dotted lines mark the stellar-mass bins used for the bin-level validation metrics in Figures 2 and 3.

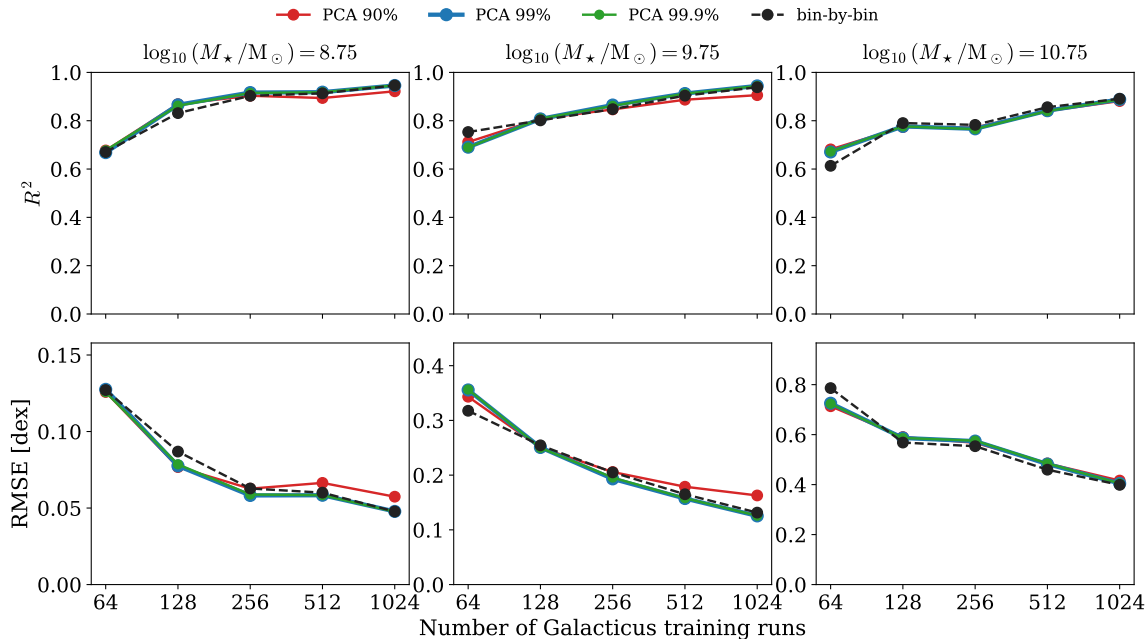


**Figure 2.** Predicted versus held-out values of the low- $z$  stellar mass function at the three stellar-mass bins highlighted in Figure 1. The  $x$ -axis values are the direct GALACTICUS predictions, while the  $y$ -axis shows the emulator predictions (with their uncertainty) at the same parameter values. The dashed lines show equality. The RMSE and  $R^2$  values quoted in each panel summarise the accuracy of these held-out predictions, using the definitions given in Section 3.2.2. For clarity, the figure shows a random 25 per cent of the held-out comparisons.

For the Sobral et al. (2013)  $H\alpha$  luminosity functions, we use the tabulated uncertainties in  $\log_{10} \Phi$  directly. The resulting uncertainties are treated as symmetric Gaussian errors in the transformed observable space. This approximation is most appropriate when the fractional uncertainties are moderate. For sparsely populated number-density bins, the underlying counting uncertainty is intrinsically asymmetric and is only approximately represented in this way.

#### 4.3. MCMC Sampling And Convergence

We sample the posterior distribution using Markov chain Monte Carlo (MCMC), implemented with the affine-invariant ensemble sampler *emcee* (Foreman-Mackey et al. 2013). Each MCMC uses 128 walkers, initialised in a small region around a point expected to lie close to the maximum of the posterior density, which was determined from a preliminary MCMC. During burn-in, the walkers disperse from this region and explore the posterior distribution. Initialising the walkers in this way helps reduce the likelihood of walkers becoming trapped around low-probability local maxima.



**Figure 3.** Held-out emulator accuracy for the  $z \simeq 0$  stellar mass function as a function of the number of GALACTICUS training runs. The top row shows the coefficient of determination,  $R^2$ , and the bottom row shows the root-mean-square error in  $\log_{10} \Phi$ . The columns correspond to the representative stellar-mass bins highlighted in Figure 1. Different curves compare PCA-GP emulators with different PCA variance thresholds to a bin-by-bin GP emulator. The improvement from smaller training subsets to the full 1024-run training set motivates the larger final campaign used for the paper calibration, while the similarity of the PCA-GP variants supports the 99 per cent PCA variance threshold used in the fiducial emulator.

For the production runs, we use 250,000 steps per walker, discarding the initial 25,000 steps as burn-in. We assess convergence using the integrated autocorrelation time,  $\tau$  (Sokal 1997), estimated separately for each sampled parameter. For the diagnostic fits used to illustrate parameter degeneracies (see Fig. 4), the post-burn-in chains extend for at least 25 times the largest measured autocorrelation time. For the final joint calibration, the chain length is 73 times the largest measured autocorrelation time. We therefore consider the chains sufficiently long for the posterior inferences presented here.

The emulator predictions are used in place of direct GALACTICUS evaluations. This dramatically speeds up inference, because calling the emulator takes a few milliseconds, while a typical GALACTICUS evaluation takes approximately 15 hours using 16 CPU cores.<sup>8</sup>

## 5. CALIBRATION RESULTS

### 5.1. Breaking Parameter Degeneracies With Multiple Observables

To build intuition for how different observables constrain the model, we consider a set of diagnostic calibrations using the stellar mass functions, the star-formation-rate function, and stellar mass–size relations. These provide a controlled example of how combining observables can reduce parameter degeneracies and reveal tensions between different aspects of the galaxy population.

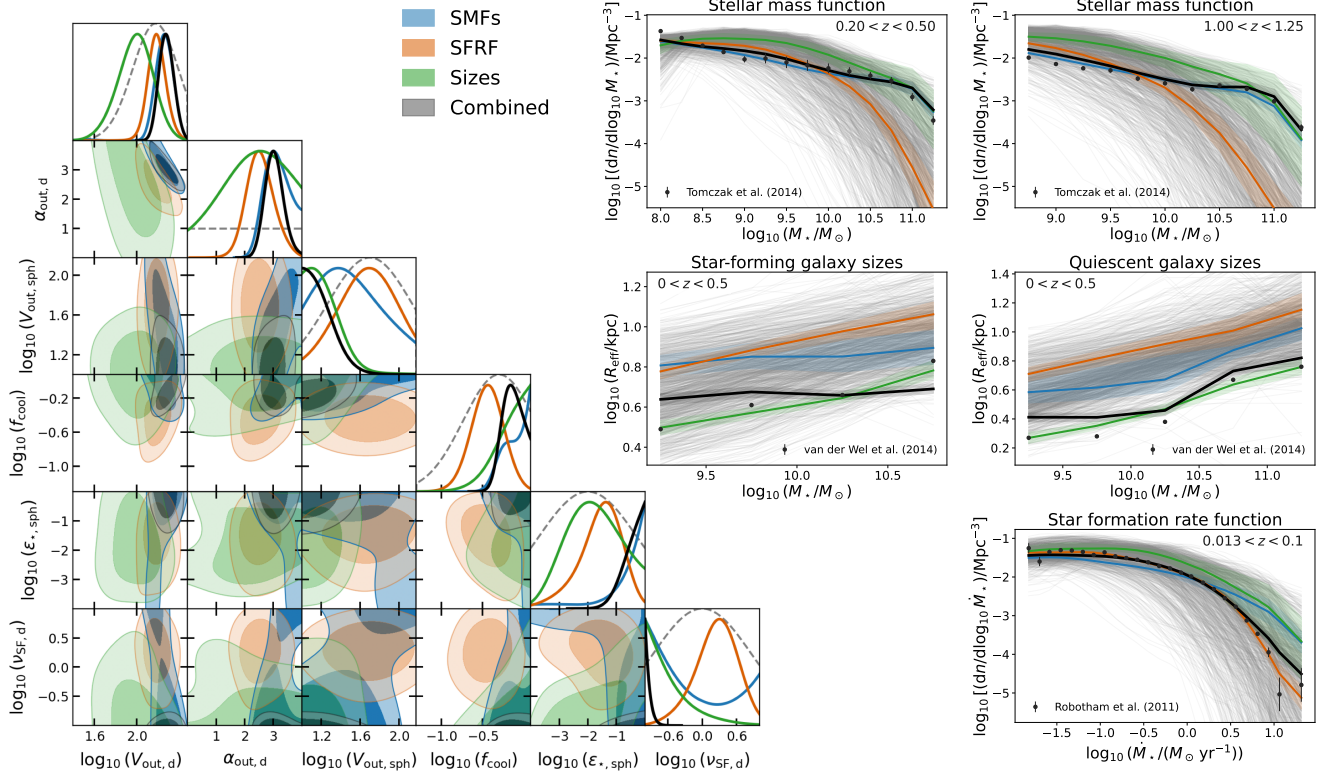
In each diagnostic fit, all 20 GALACTICUS parameters listed in Table 1 are varied. For visual clarity, Fig. 4 shows only a subset of six parameters, chosen because they control several of the main processes regulat-

ing the stellar component. These are the normalisation and circular-velocity scaling of stellar feedback in disks,  $V_{\text{out,d}}$  and  $\alpha_{\text{out,d}}$ ; the corresponding feedback velocity scale in spheroids,  $V_{\text{out,sph}}$ ; the overall gas cooling rate,  $f_{\text{cool}}$ ; and the parameters setting the rate at which gas is converted into stars in spheroids and disks,  $\epsilon_{*,\text{sph}}$  and  $\nu_{\text{SF,d}}$ . We focus on these six parameters for illustration, although the remaining parameters, including those controlling the recycling of ejected gas and the growth and feedback of black holes, also affect the stellar mass and star-formation histories.

The left-hand part of Fig. 4 shows that calibrations to individual observables leave substantial parameter degeneracies. The stellar mass functions, for example, can be reproduced by different combinations of gas cooling, star-formation efficiency, and feedback strength. They also permit a trade-off between disk and spheroid feedback, visible in the  $V_{\text{out,sph}}-V_{\text{out,d}}$  plane: stronger feedback in one component (larger  $V_{\text{out}}$ ) can be partially compensated by weaker feedback in the other. The star-formation-rate function and size–mass relations select different regions and degeneracy directions in the same parameter space. Combining the observables therefore substantially tightens the constraints on several of the parameters shown.

The size–mass relations provide a particularly clear example of both the complementarity and the tension between the observables. Over much of the prior volume, GALACTICUS produces galaxies that are too large compared with the observed relations. Matching the observed sizes favours weaker stellar feedback. This enables more effective star formation in lower-mass haloes, shifting galaxies of a given stellar mass towards less massive haloes (González et al. 2009; Lacey et al. 2016).

<sup>8</sup> Generating the 1024-run emulator training set required approximately 250,000 CPU hours in total.



**Figure 4.** An example of combining multiple observables to break degeneracies in the emulator calibration. Left: marginalised posterior constraints on a subset of six model parameters from fits to the stellar mass functions alone, the star-formation-rate function alone, galaxy sizes alone, and their combination. Diagonal panels show the one-dimensional marginalised posteriors, with grey dashed curves showing the adopted priors. Off-diagonal panels show the corresponding two-dimensional marginalised posterior distributions, with 68 and 95 per cent credible regions. Right: solid coloured curves and shaded regions show the median and central 68 per cent intervals of the emulator-predicted mean relations obtained by drawing parameters from each posterior, compared with the observational data. These intervals reflect posterior uncertainty in the calibrated model parameters and do not include observational noise or emulator uncertainty. Faint grey curves show the predictions from the 1024 training runs, illustrating the range spanned by the prior over model parameters. The combined calibration achieves reasonable agreement with all fitted observables while reducing the parameter degeneracies present in the single-observable fits.

Since galaxy sizes in semi-analytic models are linked to the sizes and angular momenta of their host dark matter haloes (e.g. Cole et al. 2000), this tends to produce smaller galaxies at fixed stellar mass.

The same shift in the relation between stellar mass and halo mass has an immediate consequence for the stellar mass functions: galaxies form more stellar mass at fixed halo mass, shifting the predicted stellar mass functions towards higher abundances. In Fig. 4, the weaker feedback favoured by the size-mass relations improves the predicted galaxy sizes, but produces stellar mass functions that lie above the observations. The combined fit therefore represents a compromise, selecting a region of parameter space that gives good simultaneous agreement with the stellar mass functions, star-formation-rate function, and galaxy sizes.

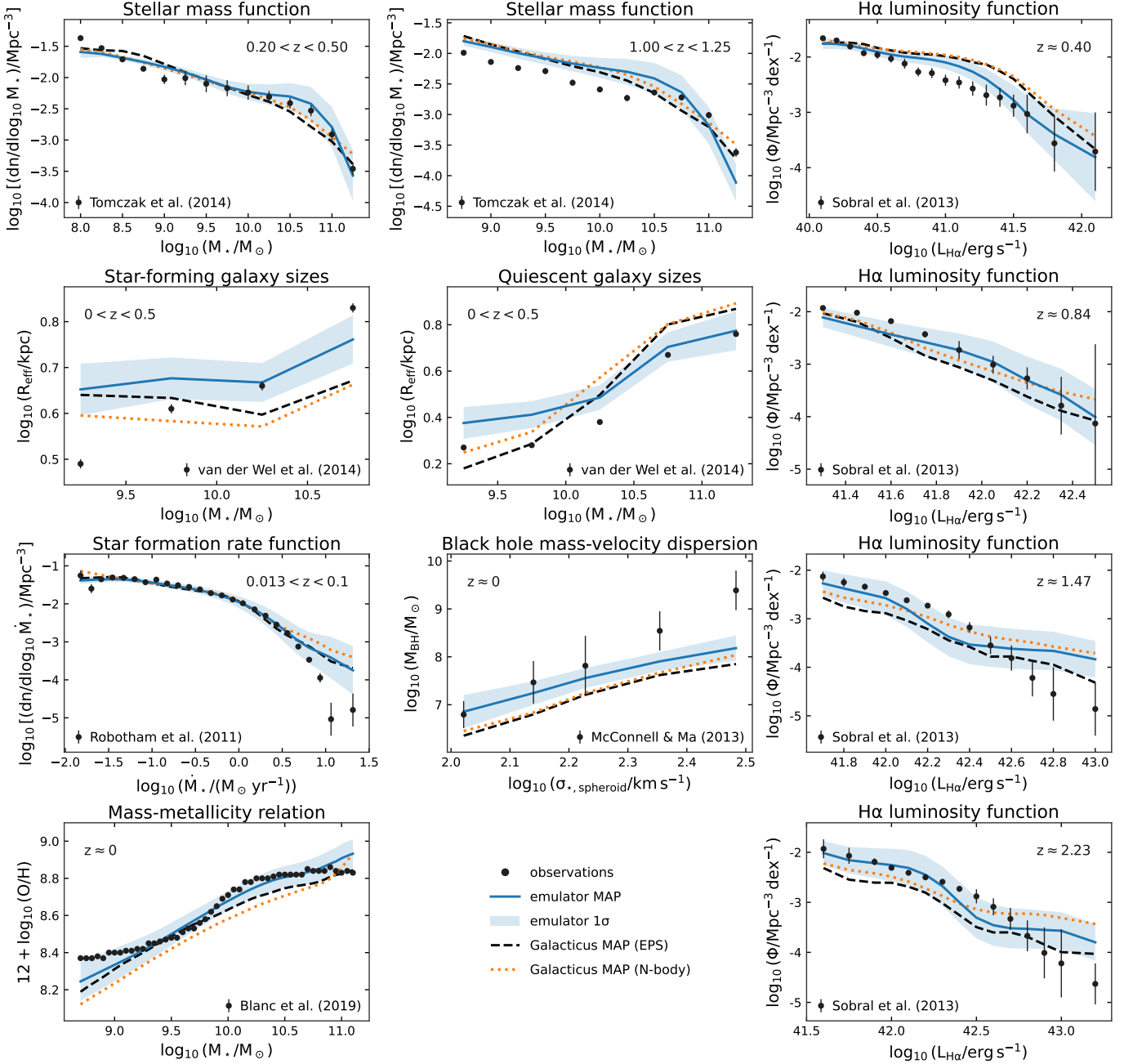
This example illustrates the benefit of calibrating to multiple observables. Different observables respond to different combinations of physical parameters, so fitting them together can constrain directions in parameter space that remain degenerate when any one observable is considered in isolation. We now apply this approach to the full set of observables used in our final calibration.

## 5.2. Final Joint Calibration

We now turn to the final calibration, in which we fit the full set of calibration observables described in Section 3.1. The posterior is sampled over the 20 GALACTICUS parameters listed in Table 1, together with the five dust-attenuation parameters defined in Appendix A.

Figure 5 shows the maximum-a-posteriori (MAP) solution from this calibration, together with direct GALACTICUS calculations with the same parameter values. We first compare the emulator prediction at the MAP point with the observational data to assess how well this solution reproduces all of the calibration observables simultaneously. The separate comparison between the emulator prediction and the direct GALACTICUS calculations tests the accuracy of the emulator at the MAP point and is discussed in Section 5.3.

The calibrated model provides a broadly successful simultaneous description of the fitted observables, although several systematic residuals remain. The low-redshift stellar mass function, the star-formation-rate function, and the gas-phase mass-metallicity relation are especially well reproduced. The higher-redshift stellar mass function shows larger residuals, with the model tending to overpredict the abundance of lower-mass galaxies. Similar behaviour has been found in comparisons of semi-analytic models, in which models calibrated



**Figure 5.** Final joint-calibration predictions for the full calibration-observable set. The left two columns show the non-H $\alpha$  calibration observables, and the right column shows the dust-attenuated H $\alpha$  luminosity functions in the four narrow redshift bins from Sobral et al. (2013) included in the likelihood. Black-points with error bars show the observational datasets listed in Table 2. The solid blue curves show the emulator prediction at the maximum-a-posteriori (MAP) parameter vector, with shaded regions indicating the emulator  $1\sigma$  predictive uncertainty. The dashed black curves show a direct GALACTICUS evaluation using the same MAP GALACTICUS parameter values on EPS merger trees; for the H $\alpha$  luminosity functions, this direct evaluation is post-processed using the MAP dust-attenuation parameters. The dotted orange curves show the corresponding direct evaluation on UNIT  $N$ -body merger trees. The final calibration provides a simultaneous fit to the stellar mass functions, size-mass relations, star-formation-rate function, black-hole scaling relation, mass-metallicity relation, and H $\alpha$  luminosity functions. The agreement between the emulator predictions and direct GALACTICUS calculations is discussed in Section 5.3.

to reproduce the low-redshift stellar mass function often overpredict the high-redshift mass function, particularly at the low-mass end (Knebe et al. 2018).

The model also predicts a black-hole mass-velocity-dispersion relation that is too shallow. As a result, the predicted black-hole masses are increasingly below the observations towards larger velocity dispersions. Fitting to this observable alone does not remove this behaviour,

suggesting that it reflects a limitation of the adopted model prescription rather than simply a trade-off introduced by the joint calibration. To determine whether the mismatch arises primarily from the black-hole masses or the predicted velocity dispersions, we also compared the direct GALACTICUS MAP calculation with the observed black-hole-bulge-mass relation of Kormendy & Ho (2013) and the empirically inferred black-hole-halo-mass

relation of Zhang et al. (2023). At the massive end, the predicted black-hole masses lie approximately 1 dex below these relations. This suggests that the discrepancy in the black-hole mass–velocity–dispersion relation primarily reflects under-massive black holes in the most massive model galaxies.

The size–mass relations also show systematic residuals, particularly at low stellar masses, where the MAP model predicts sizes that are too large for both star-forming and quiescent galaxies. As shown in Fig. 4, the observed galaxy sizes lie close to the lower envelope of the predictions spanned by the training set. Reproducing the observed sizes therefore requires the model to occupy a relatively narrow region of parameter space, making it difficult to match the sizes while simultaneously satisfying the constraints from the other observables.

For the  $H\alpha$  luminosity functions shown in the right column of Fig. 4, the MAP emulator prediction captures the broad normalisation, shape, and redshift evolution of the observed luminosity functions across the four narrow redshift bins. This broad agreement is particularly important for the *Roman* grism application, for which the abundance of emission-line galaxies is a central model prediction.

Appendix C summarises the posterior distribution of the 25 fitted parameters. Most of the marginalised posteriors retain substantial support across the regions favoured by their adopted priors. In contrast, the clearest shift is seen for the disk star-formation frequency normalisation,  $\nu_{\text{SF,d}}$ , which is constrained towards the lower edge of its prior. This preference is already apparent in the diagnostic calibrations of Fig. 4: the stellar mass functions favour low values of  $\nu_{\text{SF,d}}$ , whereas the star-formation-rate function alone favours substantially higher values.

In the adopted disk star-formation prescription, the molecular fraction is set by the Blitz & Rosolowsky (2006) pressure relation, while  $\nu_{\text{SF,d}}$  sets the rate at which molecular gas is converted into stars. For comparison, Leroy et al. (2008) found an approximately constant molecular-gas star-formation efficiency of  $5.25 \times 10^{-10} \text{ yr}^{-1}$  in nearby spiral galaxies, corresponding to a molecular-gas depletion time of approximately 1.9 Gyr. The MAP value found here,  $1.01 \times 10^{-10} \text{ yr}^{-1}$ , instead implies a depletion time of approximately 10 Gyr. The joint calibration therefore favours substantially slower conversion of molecular gas into stars than is inferred for nearby spiral galaxies.

This should not necessarily be interpreted as evidence for such long molecular-gas depletion times in real galaxies. Within the model,  $\nu_{\text{SF,d}}$  is constrained indirectly through its effects on the galaxy properties included in the calibration, and can compensate for other aspects of the gas supply and regulation of star formation. Indeed, Fig. 4 shows that different calibration observables favour different values of  $\nu_{\text{SF,d}}$ . The low value preferred by the joint calibration should therefore not be interpreted as a direct measurement of molecular-gas depletion times.

The dust-attenuation parameters are also summarised in Appendix C. Their posteriors favour a lower attenuation normalisation and a stronger stellar-mass dependence than the baseline GB10 relation. These parameters are constrained jointly with the GALACTICUS model, so their inferred values depend on the intrinsic emission-

line luminosities predicted by GALACTICUS.

### 5.3. Direct GALACTICUS Validation At The MAP Point

Figure 5 includes two direct GALACTICUS evaluations at the final MAP parameter vector. The first uses the same set of EPS merger trees as was used for the emulator training set. This provides a direct validation of the emulator prediction at the MAP point. Across the calibration observables, the direct EPS GALACTICUS calculation closely follows the emulator MAP prediction, with differences that are generally small compared with the emulator predictive uncertainty. This confirms that the emulator-derived MAP solution corresponds to a direct GALACTICUS model that provides a good simultaneous description of the calibration observables.

The same MAP parameter vector was also evaluated using  $N$ -body merger trees from one of the UNIT simulations (Chuang et al. 2019). Owing to the large volume of UNIT, we did not run GALACTICUS over the full simulation. Instead, we randomly selected 0.5 per cent of the merger trees, corresponding in number density to approximately twelve times the effective volume of the EPS training runs. The trees are selected randomly from the full simulation volume rather than from a contiguous subvolume, avoiding additional sample variance associated with the spatial clustering of haloes. For the  $H\alpha$  luminosity functions, this direct GALACTICUS run was post-processed using the MAP dust-attenuation parameters, matching the treatment of the EPS calculation.

The resulting UNIT predictions are generally close to those obtained with the EPS trees. The differences are small for most of the low-redshift observables, although a more noticeable offset is present in the mass–metallicity relation, and differences also become somewhat larger for the higher-redshift  $H\alpha$  luminosity functions. In our EPS calculation, the abundance of the  $z = 0$  host haloes is set directly by the adopted Sheth–Tormen halo mass function. At earlier redshifts, however, the halo population depends increasingly on how accurately the EPS algorithm reproduces the progenitor populations of the  $z = 0$  haloes. Comparisons with  $N$ -body simulations have shown that progenitor mass functions and halo assembly histories are sensitive to the details of the merger-tree construction (e.g. Cole et al. 2008; Parkinson et al. 2008). The somewhat larger differences seen at higher redshift are therefore not unexpected.

Overall, these tests show that the good simultaneous fit obtained at the emulator-derived MAP point is retained when the model is evaluated directly with GALACTICUS, and that applying the same calibrated parameters to UNIT  $N$ -body merger trees gives broadly similar predictions. For a future calibration intended primarily for use with  $N$ -body merger trees, it might nevertheless be preferable to construct the emulator training set using  $N$ -body trees. This consideration becomes more important as emulator uncertainties are reduced, since systematic differences associated with the merger-tree construction may then become more significant relative to the emulator uncertainty.

## 6. DISCUSSION

### 6.1. What Emulator-Assisted Calibration Enables

A central advantage of emulator-assisted calibration is that, once the emulator has been trained, the calibration can be repeated at relatively little computational cost. This makes it practical to perform separate MCMC analyses using different combinations of observables or different prior assumptions. Comparing the resulting posterior distributions can reveal which observables constrain particular parameters, which drive the model towards particular regions of parameter space, and where different observables favour incompatible parameter values. The examples in Section 5.1 illustrate how this can be used to identify degeneracies and tensions between calibration observables.

This capability becomes increasingly important as the number of varied parameters grows. In general, converged MCMC exploration becomes more demanding in higher-dimensional spaces, making repeated calibration with a computationally expensive galaxy-formation model increasingly impractical. Fast likelihood evaluations therefore make it feasible to vary a broader set of GALACTICUS parameters simultaneously. This matters because few parameters in semi-analytic models have precisely known a priori values. Holding uncertain parameters fixed makes the model more predictive, but that predictive power is conditional on assumptions about their values and may exclude otherwise viable regions of parameter space. Emulator-assisted calibration makes it practical to relax more of these assumptions and explore a larger fraction of the model’s allowed behaviour.

The emulator also provides a useful way to assess new observations. Rather than testing only whether an existing calibrated or fiducial model agrees with a new data set, one can ask whether agreement can be achieved elsewhere within the allowed model parameter space while retaining consistency with the existing calibration data. An observable that is poorly reproduced by the fiducial model might become consistent with the data after varying a particular subset of parameters. Conversely, an observable that cannot be reproduced anywhere in the region of parameter space allowed by the existing calibration provides evidence either of a limitation of the modelling framework or of tension between the observational constraints. In this way, the emulator can be used to distinguish shortcomings of a particular calibration from more fundamental tensions with the model.

The same approach could also be applied to observables directly relevant for cosmological inference. For example, by constructing training sets from a suite of  $N$ -body simulations spanning different cosmologies, one could emulate galaxy clustering statistics, such as the galaxy two-point correlation function, as functions of both cosmological and GALACTICUS parameters. Uncertainty in galaxy formation could then be marginalised over when constraining cosmology. Similar approaches have recently been demonstrated using empirical models of the galaxy–halo connection, including subhalo abundance matching and its extension to star-forming galaxies (Mahony et al. 2026; Ortega-Martinez et al. 2026). A GALACTICUS emulator would provide a route to propagating uncertainties in a physics-based model of galaxy formation into cosmological constraints.

## 6.2. Scope and Limitations

The calibration presented here should be interpreted within the fixed modelling choices adopted for this study. The emulator varies continuous parameter values, but it does not explore the full space of physical prescriptions available in GALACTICUS. Different choices for feedback, cooling, dust, environmental processes, merger treatments, or other discrete model components could introduce different parameters and change how the resulting parameter space maps onto the observables. The posterior constraints therefore describe the preferred region of the specified model family, not the full uncertainty associated with semi-analytic galaxy formation modelling.

The observational likelihood is also simplified. In particular, treating bins as independent neglects correlations that can arise from shared systematics, sample variance, calibration choices, and assumptions entering the inference of galaxy properties. Such correlations can be important when assessing agreement with a model. For example, a correlated uncertainty may allow an observable to shift coherently across several bins, whereas treating the same bins as independent would strongly penalise such a shift. Where available, full observational covariance matrices could instead be incorporated directly into the likelihood.

The finite-sampling uncertainties in the emulator training outputs may likewise be correlated between observable bins. Accounting for these correlations is possible in principle, for example through multi-output Gaussian processes (Álvarez et al. 2012) or, for the PCA emulators, by propagating the output covariance through the PCA transformation. Doing so would require an extension of the implementation used here.<sup>9</sup> The simpler treatment used here is therefore a practical approximation rather than a statement that such correlations are negligible.

## 6.3. Future Work

A clear next step is to extend the calibration to higher redshift. JWST has revealed unexpectedly abundant bright galaxy candidates at  $z \gtrsim 9$ , prompting renewed discussion of whether models calibrated primarily to lower-redshift data require changes in star-formation efficiency, burstiness, dust attenuation, or other early-Universe physics (e.g. Mirocha & Furlanetto 2023; Chemerynska et al. 2024). Emulator-assisted calibration would make it possible to ask whether a single region of GALACTICUS parameter space can simultaneously accommodate low-redshift constraints and high-redshift JWST measurements, and to identify which physical parameters are driven towards different values by the two regimes.

Future work could also use targeted or sequential training designs. The present emulator is trained over a broad prior volume, which is valuable for global exploration but becomes inefficient once the posterior is localised to a much smaller region of parameter space. Additional GALACTICUS evaluations could instead be targeted towards regions of high posterior probability or large emulator uncertainty, improving the emulator where ad-

<sup>9</sup> Our emulators use the `scikit-learn` Gaussian-process implementation, which allows heteroscedastic training uncertainties to be supplied for individual training points, but does not support a full covariance matrix for correlated training errors.

ditional accuracy matters most for the inference. Related iterative strategies have previously been used in emulator-based studies of GALFORM, where successive waves of model evaluations were concentrated within the regions of parameter space retained by *Bayesian history matching* (Bower et al. 2010; Vernon et al. 2014; Rodrigues et al. 2017). Such sequential designs may be particularly valuable in high-dimensional parameter spaces, where densely sampling the full prior volume rapidly becomes prohibitively expensive.

A longer-term goal is to extend the framework beyond parameter calibration to comparisons between alternative physical prescriptions within GALACTICUS. This would allow the data to constrain not only the values of continuous model parameters, but also choices between different treatments of the underlying galaxy-formation physics. Such an extension is more challenging because alternative prescriptions may introduce different parameters, while the Gaussian-process emulators used here are naturally formulated for interpolation over continuous parameter spaces rather than between discrete model choices. Possible approaches include training separate emulators for different model families or developing surrogate models that explicitly accommodate categorical model choices. Although technically more involved, such approaches would begin to address uncertainty in the physical prescriptions themselves, rather than only uncertainty in the continuous parameters of a fixed model family.

## 7. CONCLUSIONS

We have used Gaussian-process emulators to calibrate a GALACTICUS semi-analytic galaxy-formation model using a broad set of galaxy observables. The calibration is motivated by the need for realistic galaxy catalogues to support the design, interpretation, and analysis of the *Roman* Galaxy Redshift Survey, with emission-line luminosities predicted consistently with the other properties of the model galaxies.

We constructed a training set of 1024 full GALACTICUS calculations spanning 20 galaxy-formation parameters, and trained Gaussian-process emulators to predict the observables entering the calibration likelihood. We then used these emulators to sample the model posterior while accounting for their predictive uncertainties. The calibration simultaneously includes stellar mass functions, the star-formation-rate function, size-mass relations, the black-hole mass-velocity-dispersion relation, the gas-phase mass-metallicity relation, and dust-attenuated  $H\alpha$  luminosity functions, with five additional parameters describing the dust attenuation model. Finally, we evaluated the MAP parameter vector directly with GALACTICUS to verify the calibration obtained using the emulator.

Held-out validation shows that the differences between emulator predictions and direct GALACTICUS calculations are broadly consistent with the estimated predictive uncertainties, indicating that these uncertainties are reasonably well calibrated. We account for the emulator predictive uncertainty when comparing model predictions with the observational data, allowing the emulators to be used for posterior inference without neglecting uncertainty associated with their interpolation of the GALACTICUS calculations.

Combining multiple observables substantially reduces parameter degeneracies that arise when each observable is considered in isolation. The diagnostic calibrations also show how different observables can favour different regions of parameter space. In particular, the weaker stellar feedback favoured by the galaxy size constraints increases the predicted abundance of galaxies at fixed stellar mass, causing the stellar mass functions to lie above the observations. The joint calibration must therefore compromise between the size and abundance constraints.

The final calibration gives a broadly successful simultaneous description of the fitted galaxy population, including the abundance and redshift evolution of the  $H\alpha$  emitters that are particularly relevant to the *Roman* Galaxy Redshift Survey. A direct GALACTICUS calculation at the emulator-derived MAP point gives similar predictions, confirming that the good simultaneous fit found using the emulator is retained when the model is evaluated directly. Systematic discrepancies nevertheless remain: the model overpredicts the abundance of lower-mass galaxies in the higher-redshift stellar mass function, predicts sizes that are too large at low stellar mass, and produces black holes that are under-massive in the most massive galaxies. These residuals provide useful targets for future development of GALACTICUS.

Applying the same calibrated parameters to merger trees extracted from a UNIT  $N$ -body simulation produces broadly similar predictions to those obtained with the EPS trees used for emulator training. This supports applying the calibrated model to  $N$ -body merger trees more generally. Differences between the EPS- and  $N$ -body-based predictions remain, particularly for some higher-redshift observables, suggesting that future calibrations may benefit from constructing the emulator training set directly from  $N$ -body merger trees, especially if emulator uncertainties are reduced.

The calibrated model provides a practical basis for constructing realistic mock galaxy populations to support the design, interpretation, and analysis of the *Roman* Galaxy Redshift Survey. More generally, our results demonstrate that emulation makes posterior calibration against many complementary observables practical for galaxy-formation models that would otherwise be too computationally expensive to explore directly. The same approach can be applied with different calibration datasets, model prescriptions, or target applications.

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(Jones et al. 2001).

OpenAI's Codex<sup>10</sup> was used to assist with coding the emulator and with manuscript preparation. The emulator implementation was reviewed by the authors and validated using the cross-validation and held-out-GALACTICUS tests described in Section 3.2.2 and Appendix B.1. All AI-assisted output was reviewed, edited, and verified by the authors, who take full responsibility for the content of the manuscript.

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<sup>10</sup> <https://openai.com/codex/>

## APPENDIX

## A. EMISSION-LINE DUST MODEL DETAILS

This appendix describes the dust attenuation applied to the emission-line luminosities used in the  $H\alpha$  luminosity-function calibration. The intrinsic line luminosities are calculated within GALACTICUS, with the emission-line model described in detail by Weerasooriya et al. (in preparation). For star-forming regions, the model uses pre-computed CLOUDY photoionization calculations, as described in Section 2.5. AGN emission lines are calculated using a separate CLOUDY tabulation with spectra appropriate for AGN, with the luminosity of the ionizing source determined from the black hole accretion rate and the GALACTICUS accretion-disk model. We sum the contributions from star-forming regions in the disk and spheroid together with those from the AGN to obtain the intrinsic line luminosities. These are attenuated using the model below before constructing luminosity functions in the Sobral et al. (2013) bins.

The baseline attenuation model is the local  $H\alpha$  attenuation–stellar-mass relation of GB10. Defining

$$X = \log_{10} \left( \frac{M_{\star}}{10^{10} M_{\odot}} \right), \quad (\text{A1})$$

the baseline attenuation is

$$A_{H\alpha}^{\text{GB10}} = 0.91 + 0.77X + 0.11X^2 - 0.09X^3. \quad (\text{A2})$$

To avoid extrapolating this empirical relation far beyond the stellar-mass range over which it was calibrated, we clip the stellar mass entering Equation A2 to the range  $10^8 < M_{\star}/M_{\odot} < 10^{11}$  before evaluating  $X$ .

We allow the attenuation relation to depart from this baseline according to

$$\bar{A}_{H\alpha}(M_{\star}, z) = A_{H\alpha}^{\text{GB10}}(M_{\star}) + \delta_0 + \delta_M X + \delta_z u + \delta_{Mz} Xu, \quad (\text{A3})$$

where

$$u = \ln \left( \frac{1+z}{1+z_{\text{piv}}} \right), \quad (\text{A4})$$

and we adopt  $z_{\text{piv}} = 1$ . The parameters  $\delta_0$  and  $\delta_M$  therefore allow departures from the normalization and stellar-mass dependence of the GB10 relation at the pivot redshift, while  $\delta_z$  and  $\delta_{Mz}$  allow these departures to evolve with redshift.

For an individual galaxy, the attenuation is described by

$$A_{H\alpha}(M_{\star}, z) = \bar{A}_{H\alpha}(M_{\star}, z) + \epsilon_A, \quad \epsilon_A \sim \mathcal{N}(0, \sigma_A^2), \quad (\text{A5})$$

with negative values truncated at  $A_{H\alpha} = 0$ .

The five dust parameters sampled in the final calibration are  $\delta_0$ ,  $\delta_M$ ,  $\delta_z$ ,  $\delta_{Mz}$ , and  $\sigma_A$ . We adopt independent normal priors with mean zero and standard deviation 0.5 for the four shift parameters. For  $\sigma_A$ , we adopt a normal prior with mean 0.25 and standard deviation 0.1, truncated at  $\sigma_A = 0$ . These priors allow departures from the local GB10 relation while retaining it as the reference model for nebular attenuation.

The attenuated  $H\alpha$  luminosity is

$$\log_{10} L_{H\alpha}^{\text{obs}} = \log_{10} L_{H\alpha}^{\text{int}} - 0.4 A_{H\alpha}. \quad (\text{A6})$$

Rather than drawing a stochastic attenuation for each galaxy when  $\sigma_A > 0$ , we compute the expected contribution of each galaxy to each luminosity bin by integrating over the attenuation distribution, including the truncation at  $A_{H\alpha} = 0$ . This avoids introducing Monte Carlo noise into the luminosity functions while retaining the redistribution of galaxies between luminosity bins caused by attenuation scatter.

For each of the 1024 GALACTICUS training points, we independently draw one set of dust parameters from these priors, giving 1024 training points distributed over the combined 25-dimensional parameter space. Because the dust attenuation is applied in post-processing, additional dust realizations can be generated at negligible cost compared with additional GALACTICUS runs. We tested training the emulator with multiple dust realizations per GALACTICUS run, but found only modest improvements in held-out predictive accuracy. We therefore use one dust realization per run for the fiducial emulator, keeping the training set smaller and reducing the computational cost of emulator training and evaluation. The  $H\alpha$  luminosity-function emulator consequently has 25 input dimensions: the 20 GALACTICUS parameters and the five dust parameters.

## B. OBSERVABLE AND EMULATOR DETAILS

## B.1. Held-Out Emulator Validation Summary

The RMSE and  $R^2$  statistics defined in Section 3.2.2 quantify the agreement between the emulator mean and the direct GALACTICUS calculation. They are useful descriptive measures of predictive accuracy, and are used in Figures 2

**Table 3**

Five-fold held-out validation summary for the calibration-observable emulators trained on the full 1024-run campaign. RMSE and  $R^2$  are defined in Section 3.2.2. For each held-out prediction, the standardized residual  $z_n$  is normalized by the combined emulator predictive uncertainty and finite-sampling uncertainty of the held-out GALACTICUS calculation, as defined in Appendix B.1. The entries are medians over the bins of each observable.  $f_{1\sigma}$  and  $f_{2\sigma}$  are the fractions of held-out predictions with  $|z_n| \leq 1$  and  $|z_n| \leq 2$ , respectively.

| Calibration observable                  | Redshift          | RMSE | $R^2$ | RMS( $z$ ) | $f_{1\sigma}$ | $f_{2\sigma}$ |
|-----------------------------------------|-------------------|------|-------|------------|---------------|---------------|
| Stellar mass function                   | $0.20 < z < 0.50$ | 0.11 | 0.92  | 1.17       | 0.69          | 0.94          |
|                                         | $1.00 < z < 1.25$ | 0.24 | 0.92  | 1.14       | 0.66          | 0.92          |
| Star-formation-rate function            | $0.013 < z < 0.1$ | 0.11 | 0.88  | 1.08       | 0.70          | 0.94          |
| Size–mass relation (star-forming)       | $0 < z < 0.5$     | 0.05 | 0.90  | 1.14       | 0.69          | 0.93          |
| Size–mass relation (quiescent)          | $0 < z < 0.5$     | 0.06 | 0.84  | 1.13       | 0.70          | 0.93          |
| Black-hole–velocity–dispersion relation | $z \approx 0$     | 0.32 | 0.79  | 1.40       | 0.64          | 0.90          |
| Gas-phase mass–metallicity relation     | $z \approx 0$     | 0.08 | 0.89  | 1.18       | 0.74          | 0.94          |
|                                         | $z = 0.40$        | 0.14 | 0.85  | 1.11       | 0.67          | 0.94          |
|                                         | $z = 0.84$        | 0.60 | 0.78  | 1.11       | 0.67          | 0.92          |
|                                         | $z = 1.47$        | 0.74 | 0.78  | 1.05       | 0.72          | 0.94          |
| $H\alpha$ luminosity function           | $z = 2.23$        | 0.79 | 0.78  | 1.05       | 0.73          | 0.93          |

and 3. However, neither statistic isolates the interpolation error of the emulator. Each held-out GALACTICUS prediction is measured from a finite simulated galaxy population and is therefore itself a noisy estimate of the underlying model prediction.

For a given observable bin, let  $g_n$  be the value measured from held-out GALACTICUS calculation  $n$ , and let  $f_n$  denote the corresponding infinite-sampling model prediction, i.e. the expectation that would be approached as the number of simulated haloes becomes arbitrarily large. We write

$$g_n = f_n + \epsilon_n, \quad \text{Var}(\epsilon_n) = \sigma_{\text{sim},n}^2, \quad (\text{B1})$$

where  $\sigma_{\text{sim},n}$  is the estimated finite-sampling uncertainty of the held-out calculation. This is the same type of uncertainty that is supplied to the GP as a training-output uncertainty when building the emulators.

Let  $m_n$  be the emulator mean at the held-out parameter point and  $\sigma_{\text{emu},n}$  its predictive uncertainty for the latent model prediction. Treating the GP predictive uncertainty and finite-sampling uncertainty as independent, the expected variance of the residual,  $m_n - g_n$ , is

$$s_n^2 = \sigma_{\text{emu},n}^2 + \sigma_{\text{sim},n}^2. \quad (\text{B2})$$

We therefore define the standardized residual

$$z_n = \frac{m_n - g_n}{s_n}. \quad (\text{B3})$$

For each observable bin, we summarize these residuals using

$$\text{RMS}(z) = \left[ \frac{1}{N_{\text{val}}} \sum_n z_n^2 \right]^{1/2}, \quad (\text{B4})$$

and the coverage fractions

$$f_{k\sigma} = \frac{1}{N_{\text{val}}} \sum_n \mathbb{I}(|z_n| \leq k), \quad k \in \{1, 2\}, \quad (\text{B5})$$

where  $\mathbb{I}$  is the indicator function. If the combined uncertainties are correctly calibrated and the standardized residuals are approximately Gaussian, we expect  $\text{RMS}(z) \approx 1$ ,  $f_{1\sigma} \approx 0.68$ , and  $f_{2\sigma} \approx 0.95$ .

Table 3 gives the median value of each diagnostic over the bins of an observable. Once the finite-sampling uncertainty of the held-out calculations is included, the coverage of the  $H\alpha$  luminosity-function emulators is reasonably close to the Gaussian expectation at all four redshifts. In particular, the larger raw RMSE values at higher redshift do not correspond to a comparable deterioration in  $\text{RMS}(z)$ .

This distinction is especially important for the  $H\alpha$  luminosity functions. For a survey covering a fixed area, the comoving volume probed by the observational samples generally increases towards higher redshift, allowing the observed luminosity functions to extend to lower number densities. By contrast, every GALACTICUS calculation used here has the same effective comoving volume. The higher-redshift luminosity-function bins therefore contain fewer simulated galaxies and have larger finite-sampling uncertainties. The direct held-out calculations consequently become noisier with increasing redshift, even if the accuracy of the emulator for the underlying model prediction remains similar.

### B.2. Treatment of floored and missing emulator training targets

The transformations described in Section 3.1.1 require a finite emulator target for every training model and observable bin. This appendix specifies how model predictions that did not satisfy this requirement were treated. These operations

**Table 4**

Floored or median-replaced predictions in the production emulator training set. Each observable has 1024 training predictions per bin. The affected region gives the coordinates of bins in which at least one prediction required treatment, while the affected-points fraction is calculated over all training models and bins for the corresponding observable.

| Calibration observable                  | Treatment          | Affected bins | Affected region                                                  | Affected fraction |
|-----------------------------------------|--------------------|---------------|------------------------------------------------------------------|-------------------|
| Black-hole–velocity–dispersion relation | Median replacement | 2/5           | $\sigma_* \geq 226 \text{ km s}^{-1}$                            | 8.0%              |
| Stellar mass function                   | Floor              | 5/14          | $M_* \geq 10^{10.25} M_\odot, 0.2 < z < 0.5$                     | 3.9%              |
|                                         |                    | 6/11          | $M_* \geq 10^{10.00} M_\odot, 1 < z < 1.25$                      | 8.9%              |
| Gas-phase mass–metallicity relation     | Median replacement | 18/49         | $M_* \geq 10^{10.25} M_\odot$                                    | 1.1%              |
| Star-formation-rate function            | Floor              | 8/25          | $\dot{M}_* \gtrsim 2.0 M_\odot \text{ yr}^{-1}$                  | 3.0%              |
| Size–mass relation (star-forming)       | Median replacement | 1/4           | $M_* = 10^{10.75} M_\odot$                                       | 0.1%              |
| Size–mass relation (quiescent)          | Median replacement | 2/5           | $M_* \geq 10^{10.75} M_\odot$                                    | 1.4%              |
|                                         |                    | 5/18          | $L_{\text{H}\alpha} \geq 10^{41.4} \text{ erg s}^{-1}, z = 0.40$ | 0.7%              |
|                                         |                    | 7/9           | $L_{\text{H}\alpha} \geq 10^{41.6} \text{ erg s}^{-1}, z = 0.84$ | 9.9%              |
|                                         |                    | 13/13         | All bins, $z = 1.47$                                             | 26.0%             |
| H $\alpha$ luminosity function          | Floor              | 15/15         | All bins, $z = 2.23$                                             | 30.0%             |

apply only to the model predictions used to train the emulators; no values in the observational calibration data are replaced or modified.

#### B.2.1. Number-density observables

For the stellar-mass and star-formation-rate functions, predictions at or below a minimum number density were assigned

$$\log_{10} \left( \frac{\Phi_{\text{floor}}}{\text{Mpc}^{-3} \text{ dex}^{-1}} \right) = -6.64.$$

For the dust-attenuated H $\alpha$  luminosity functions, the corresponding floor was

$$\log_{10} \left( \frac{\Phi_{\text{floor}, \text{H}\alpha}}{\text{Mpc}^{-3} \text{ dex}^{-1}} \right) = -8.$$

Floored stellar-mass- and star-formation-rate-function predictions were assigned a training uncertainty of 0.5 dex, while floored H $\alpha$  luminosity functions were assigned a training uncertainty of 2 dex. These enlarged uncertainties reduce the influence of floored predictions on the emulator training.

Both floors lie well below the observational data entering the final calibration. The smallest observed number density is a factor of 40 above the corresponding floor, for the star-formation-rate function, while the stellar-mass and H $\alpha$  luminosity functions remain at least three orders of magnitude above their floors.

#### B.2.2. Undefined binned relations

For binned relations, the predicted ordinate is undefined if a model produces no galaxies satisfying the selection for a particular bin. Such entries must nevertheless be assigned finite values because the PCA and Gaussian-process training require a complete data vector for every training model. This affected the emulated ordinates  $\log_{10} R_{\text{eff}}$ ,  $\log_{10} M_{\text{BH}}$ , and  $12 + \log_{10}(\text{O}/\text{H})$ .

Each undefined entry was replaced by the median of the valid training-set predictions in the same bin, with the replacement therefore determined independently for each bin. Each replaced entry was assigned a training uncertainty of 5 in the units of the emulated ordinate, so that it contributes little information to the emulator training.

#### B.2.3. Incidence in the production training set

Table 4 reports the incidence of flooring and median replacement among the 1024 production training models. A bin is listed as affected if at least one training model required treatment. The affected fraction is the fraction of all training-model–bin combinations for the corresponding table row that required treatment.

For the standard observables, treated entries occur predominantly in bins selecting massive galaxies or high star-formation rates, where some training models contain few or no qualifying galaxies. The incidence is substantially larger for the two highest-redshift H $\alpha$  luminosity functions. At these redshifts, the fixed effective volume of each GALACTICUS calculation contains fewer galaxies in many of the luminosity bins probed by the observational data. In addition, some regions of the dust-model parameter space attenuate the H $\alpha$  emission sufficiently strongly that a model produces no emitters in a bin, including in some relatively faint luminosity bins.

Both treatments predominantly affect regions of parameter space that produce galaxy populations inconsistent with the calibration data. A number-density prediction below the adopted floor is already far below the corresponding observed abundance. Similarly, an undefined binned relation usually occurs because a model produces no galaxies satisfying the relevant selection, typically in a massive or otherwise sparsely populated part of the galaxy population. Such models will generally also be disfavoured by the number-density observables that constrain the abundance of

these galaxies. The precise treatment of these predictions is therefore less important than the accuracy of the emulator in regions of parameter space that provide plausible fits to the calibration data.

### B.3. Covariance And Uncertainty Treatment

The observable vectors measured from the finite galaxy populations in each GALACTICUS training calculation have finite-sampling uncertainties. These uncertainties are estimated separately for each training model and observable bin and are included when training the Gaussian-process emulators. Here we describe how the per-bin training uncertainties are propagated through the standardisation and PCA transformations and how the resulting GP predictive uncertainties are reconstructed in observable space.

For training model  $n$ , let  $y_{nb}$  denote the prediction in observable bin  $b$ , after applying the observable-space transformation described in Section 3.1.1, and let  $\sigma_{nb}^2$  denote the estimated finite-sampling variance of that prediction in the same transformed space. We retain only the diagonal per-bin variances, neglecting correlations in the finite-sampling fluctuations between different observable bins.

For the PCA emulators, each observable bin is first standardised using its training-set mean,  $\mu_b$ , and standard deviation,  $s_b$ , as described in Section 3.1.2:

$$z_{nb} = \frac{y_{nb} - \mu_b}{s_b}. \quad (\text{B6})$$

The corresponding finite-sampling variance is

$$\text{Var}(z_{nb}) = \frac{\sigma_{nb}^2}{s_b^2}. \quad (\text{B7})$$

The coefficient of training model  $n$  along PCA component  $k$  is

$$a_{nk} = \sum_b e_{kb} z_{nb}, \quad (\text{B8})$$

where  $e_{kb}$  is the  $b$ th element of the  $k$ th PCA component. Under the diagonal approximation for the per-bin training covariance, the finite-sampling variance of this coefficient is

$$\sigma_{a,nk}^2 = \sum_b e_{kb}^2 \frac{\sigma_{nb}^2}{s_b^2}. \quad (\text{B9})$$

Although the PCA transformation can induce covariance between the finite-sampling uncertainties of different PCA coefficients, these cross-component covariances are not retained; each coefficient is modelled with an independent scalar GP.

Before fitting its GP, each retained PCA coefficient is itself standardised:

$$\tilde{a}_{nk} = \frac{a_{nk} - \bar{a}_k}{q_k}, \quad (\text{B10})$$

where  $\bar{a}_k$  and  $q_k$  are the training-set mean and standard deviation of coefficient  $k$ . The training-output variance supplied to the GP for this training point is therefore

$$\sigma_{\tilde{a},nk}^2 = \frac{\sigma_{a,nk}^2}{q_k^2}. \quad (\text{B11})$$

These variances are added to the diagonal of the covariance matrix obtained by evaluating the GP kernel at the training points. The resulting training-noise term is heteroscedastic: it can differ between training models and between PCA coefficients.

For the bin-by-bin emulators used in the validation comparisons, the same procedure is applied without the PCA transformation. The finite-sampling variance in each bin is propagated through the standardisation of the corresponding scalar GP target and added to the diagonal of its training covariance matrix.

At a new point in parameter space, let  $\sigma_{a,k}^2(\boldsymbol{\theta})$  denote the predictive variance from the GP for PCA coefficient  $k$ , after reversing the standardisation of that coefficient. Treating the retained PCA-coefficient predictions as independent, the marginal emulator variance in observable bin  $b$  is

$$\sigma_{\text{emu},b}^2(\boldsymbol{\theta}) = s_b^2 \sum_{k=1}^K e_{kb}^2 \sigma_{a,k}^2(\boldsymbol{\theta}). \quad (\text{B12})$$

The PCA reconstruction also induces covariance between different observable bins, but only the marginal variance of each bin is retained. These diagonal emulator variances are used in the calibration likelihood described in Section 4.2.

The marginal predictive variances are therefore obtained by propagating the uncertainties of the retained PCA coefficients through the PCA reconstruction. The adequacy of the 99 per cent variance threshold, including any error introduced by the compression, is assessed empirically through the held-out validation tests in Section 3.2.2.

**Table 5**

Posterior summary for the final calibrated model. The columns  $p_{16}$ ,  $p_{50}$ , and  $p_{84}$  give the 16th, 50th, and 84th percentiles of the marginalized posterior. The  $q$  columns give the corresponding locations in the adopted prior, expressed as percentiles on a 0–100 scale. Units for dimensional parameters are shown in the parameter column.

| Parameter                              | MAP     | Posterior |          |          | Prior percentile |          |          |          |
|----------------------------------------|---------|-----------|----------|----------|------------------|----------|----------|----------|
|                                        |         | $p_{16}$  | $p_{50}$ | $p_{84}$ | $q_{\text{MAP}}$ | $q_{16}$ | $q_{50}$ | $q_{84}$ |
| Star formation and stellar populations |         |           |          |          |                  |          |          |          |
| $\nu_{\text{SF,d}}/\text{Gyr}^{-1}$    | 0.101   | 0.102     | 0.107    | 0.119    | 0.1              | 0.3      | 1.0      | 2.5      |
| $\epsilon_{*,\text{sph}}$              | 0.0839  | 0.0860    | 0.108    | 0.135    | 77.0             | 77.3     | 79.9     | 82.3     |
| $y_z$                                  | 0.0362  | 0.0328    | 0.0355   | 0.0382   | 95.9             | 90.6     | 94.9     | 98.2     |
| Stellar feedback                       |         |           |          |          |                  |          |          |          |
| $V_{\text{out,d}}/\text{km s}^{-1}$    | 170     | 155       | 167      | 183      | 65.2             | 57.6     | 63.6     | 71.3     |
| $\alpha_{\text{out,d}}$                | 2.55    | 2.37      | 2.57     | 2.73     | 63.7             | 59.1     | 64.2     | 68.3     |
| $V_{\text{out,sph}}/\text{km s}^{-1}$  | 10.8    | 10.5      | 11.9     | 14.6     | 1.0              | 0.7      | 2.8      | 6.9      |
| Gas cooling and recycling              |         |           |          |          |                  |          |          |          |
| $f_{\text{cool}}$                      | 0.436   | 0.395     | 0.447    | 0.526    | 54.4             | 49.6     | 55.7     | 63.8     |
| $r_{\text{core}}/R_{\text{vir}}$       | 0.303   | 0.258     | 0.330    | 0.456    | 56.4             | 49.2     | 60.2     | 74.4     |
| $\gamma_{\text{reinc}}$                | 1.64    | 1.64      | 2.19     | 2.82     | 1.2              | 1.2      | 4.8      | 12.6     |
| $\delta_{\text{reinc},1}$              | −0.127  | −0.935    | −0.450   | −0.0224  | 31.2             | 17.8     | 25.8     | 33.0     |
| $\delta_{\text{reinc},2}$              | 3.60    | 2.95      | 3.81     | 4.50     | 76.7             | 65.9     | 80.2     | 91.6     |
| Mergers and disk instabilities         |         |           |          |          |                  |          |          |          |
| $f_{\text{major}}$                     | 0.467   | 0.344     | 0.422    | 0.476    | 97.4             | 79.7     | 92.7     | 98.2     |
| $f_{\text{orbit}}$                     | 1.52    | 1.48      | 1.94     | 2.62     | 80.1             | 78.7     | 90.9     | 97.6     |
| $f_{j,\text{sph}}^{\text{bar}}$        | 0.177   | 0.105     | 0.157    | 0.236    | 40.3             | 9.6      | 31.3     | 62.8     |
| Supermassive black holes               |         |           |          |          |                  |          |          |          |
| $M_{\text{BH,seed}}/(10^4 M_{\odot})$  | 7.17    | 1.68      | 4.76     | 8.21     | 96.2             | 76.6     | 91.1     | 97.8     |
| $\epsilon_{\text{BH,radio}}$           | 0.183   | 0.171     | 0.268    | 0.419    | 16.2             | 13.0     | 41.4     | 75.4     |
| $\epsilon_{\text{BH,wind}}$            | 0.00321 | 0.00150   | 0.00404  | 0.00937  | 61.7             | 31.6     | 70.3     | 92.2     |
| $\alpha_{\text{Bondi,sph}}$            | 0.0524  | 0.149     | 0.442    | 1.26     | 0.1              | 3.1      | 11.0     | 25.5     |
| $\alpha_{\text{Bondi,hot}}$            | 3.29    | 3.26      | 15.3     | 49.2     | 41.0             | 40.8     | 73.7     | 92.7     |
| $x_{\text{thin,max}}$                  | 21.8    | 16.9      | 23.1     | 27.9     | 95.8             | 92.1     | 96.6     | 99.1     |
| Dust attenuation                       |         |           |          |          |                  |          |          |          |
| $\delta_0$                             | −0.738  | −0.886    | −0.688   | −0.279   | 7.0              | 3.8      | 8.4      | 28.9     |
| $\delta_M$                             | 0.795   | 0.0575    | 0.711    | 1.07     | 94.4             | 54.6     | 92.2     | 98.4     |
| $\delta_z$                             | 0.518   | −0.525    | 0.146    | 0.605    | 85.0             | 14.7     | 61.5     | 88.7     |
| $\delta_{Mz}$                          | 0.0474  | −0.155    | 0.433    | 1.15     | 53.8             | 37.9     | 80.7     | 98.9     |
| $\sigma_A$                             | 0.258   | 0.208     | 0.290    | 0.375    | 53.0             | 33.5     | 65.5     | 89.3     |

### C. CALIBRATED MODEL POSTERIOR

Table 5 summarizes the MAP point and marginalized posterior distribution for the final calibration. In addition to the parameter values themselves, we report their locations within the adopted priors, making it easier to identify parameters for which the calibration favours values in the tails of their prior distributions. As discussed in Section 2.3, the priors are generally centred on values used in previous GALACTICUS configurations or suggested by earlier calibration experiments, rather than representing precise prior constraints on the underlying physical parameters.

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**Figure 6.** Marginalised posterior distributions for the 20 GALACTICUS parameters in the final joint calibration. The inset shows the corresponding corner plot for the five dust-attenuation parameters. Diagonal panels show the one-dimensional marginalised posteriors, with dashed grey curves showing the adopted priors. Dashed blue vertical and horizontal lines mark the maximum-a-posteriori parameter values. Off-diagonal panels show the corresponding two-dimensional marginalised posterior distributions, with 68 and 95 per cent credible regions.